

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

MATH 330: MARTINGALES AND BROWNIAN MOTION

INSTRUCTOR: PROFESSOR ROBERT DALANG



Course Notes

Spring 2025

Transcribed in $\text{\LaTeX}$ by Cedric Phillips

Original notes template by LazzzzyJack

EPFL

Contents

1	Introduction	2
1.1	Conditional expectation	2
1.2	Martingales	5
1.3	Super- and submartingales	9
1.4	Stopping	11
2	Martingale theory	21
2.1	Martingale convergence theorems	21
2.2	Branching Processes (Galton-Watson Processes)	28
2.2.1	Hypotheses A and B	28
2.2.2	Asymptotic behaviour of X_n as $n \rightarrow \infty$	32
3	Brownian motion	34
3.1	Behaviours of Brownian motion	36
3.2	Transformations of Brownian motion	39
3.3	Exit times	43
3.4	Zero sets	46
3.5	Hausdorff dimension	48
3.6	Continuity of sample paths	51

Chapter 1: Introduction

Example 1.0.1

To motivate. A player plays a sequence of fair games (of chance). At each stage, the expected reward, given all that has been previously observed, is zero.

Let X_n be the cumulative gain or loss after n games. Can the player guarantee that his expected reward when he leaves the game be positive? Can he find a random time τ (not decided in advance) such that $\mathbb{E}(X_\tau) > 0$?

We will see that for "reasonable" τ , $\mathbb{E}(X_\tau) = 0$. If $\mathbb{E}(X_\tau) > 0$, then τ will usually correspond to a strategy that is problematic.

1.1 Conditional expectation

Remark.

Elementary conditional probability:

$$\mathbb{P}(G|H) = \frac{\mathbb{P}(G \cap H)}{\mathbb{P}(H)} \quad \text{if } \mathbb{P}(H) > 0$$

Proposition 1.1.1

If an r.v X is of the discrete type, i.e., X takes values in $\{x_1, x_2, \dots, x_n\} \subset \mathbb{R}$, then $\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i)$. But this only holds if $\sum_i |x_i| \mathbb{P}(X = x_i) < \infty$.

If $\mathbb{E}(|X|) < \infty$, then X is said to be integrable.

Now consider a function $h : \mathbb{R} \rightarrow \mathbb{R}$.

Then $\mathbb{E}(h(X)) = \sum_i h(x_i) \mathbb{P}(X = x_i)$, but only if $\sum_i |h(x_i)| \mathbb{P}(X = x_i) < \infty$.

Proposition 1.1.2

Now let X be a continuous random variable, so it has a probability density function $f(x)$. Then $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx$ (for $a \leq b$).

Then $\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx$, but only if $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, i.e., $\mathbb{E}(|X|) < \infty$.

Now consider a function $h : \mathbb{R} \rightarrow \mathbb{R}$.

Then $\mathbb{E}(h(X)) = \int_{-\infty}^{\infty} h(x)f(x)dx$; likewise only holds if $\int_{-\infty}^{\infty} |h(x)|f(x)dx < \infty$.

Definition 1.1.3: Conditional expectation

Let (X, Y) be a random vector in $\mathbb{R} \times \mathbb{R}^m$, such that $\mathbb{E}(|X|) < \infty$. Then $\mathbb{E}(X|Y)$ is the integrable r.v of the form $\varphi(Y)$, where $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}$, such that, for all bounded $h : \mathbb{R}^m \rightarrow \mathbb{R}$:

$$\mathbb{E}(\mathbb{E}(X|Y) \cdot h(Y)) = \mathbb{E}(X \cdot h(Y))$$

If $\mathbb{E}(X^2) < \infty$, then $\mathbb{E}(X|Y) = \varphi(Y)$ is such that $\mathbb{E}((X - \varphi(Y))^2) \leq \mathbb{E}((X - \psi(Y))^2)$ for all $\psi : \mathbb{R}^m \rightarrow \mathbb{R}$.

Proposition 1.1.4

Formulas for $\mathbb{E}(X|Y)$ (discrete case).

Let (X, Y) be a discrete r.v with $\mathbb{E}(|X|) < \infty$. Then:

$$\text{a) Set } \mathbb{E}(X|Y = y) = \sum_x x\mathbb{P}(X = x|Y = y) = \frac{1}{\mathbb{P}(Y = y)} \sum_x x\mathbb{P}(X = x, Y = y)$$

(if $\mathbb{P}(Y = y) > 0$.)

b) Define a function φ by $\varphi(y) = \mathbb{E}(X|Y = y)$. Then $\mathbb{E}(X|Y) = \varphi(Y)$.

Proposition 1.1.5

Formulas for $\mathbb{E}(X|Y)$ (continuous case).

Let (X, Y) be a continuous r.v with values in $\mathbb{R} \times \mathbb{R}^m$, with joint density $f : \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$ given by:

$$\mathbb{P}(X \in A, Y \in B) = \int_A dx \int_B dy \cdot f(x, y)$$

(for $A \subset \mathbb{R}, B \subset \mathbb{R}^m$).

Now recall the marginal density of Y : $f_Y(y) = \int_{\mathbb{R}} dx \cdot f(x, y)$. Assume $\mathbb{E}(|X|) < \infty$.

a) Set $\mathbb{E}(X|Y = y) = \int_{\mathbb{R}} x \frac{f(x, y)}{f_Y(y)} dx \quad (f_Y(y) > 0)$

b) Define a function φ by $\varphi(y) = \mathbb{E}(X|Y = y)$. Then $\mathbb{E}(X|Y) = \varphi(Y)$.

c) Let $h : \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$; then $\mathbb{E}(h(X, Y)|Y = y) = \int_{\mathbb{R}} dx \cdot h(x, y) \frac{f(x, y)}{f_Y(y)}$

d) Define a function ψ by $\psi(y) = \mathbb{E}(h(X, Y)|Y = y)$ then $\mathbb{E}(h(X, Y)|Y) = \psi(Y)$.

Proposition 1.1.6

Properties of conditional expectation:

a) Linearity: $\mathbb{E}(\alpha X + \beta Z|Y) = \alpha \mathbb{E}(X|Y) + \beta \mathbb{E}(Z|Y)$.

b) Monotonicity: If $X \leq Z$, then $\mathbb{E}(X|Y) \leq \mathbb{E}(Z|Y)$.

c) Iteration: $\mathbb{E}(\mathbb{E}(X|Y, Z)|Y) = \mathbb{E}(X|Y)$

d) Composition: $\mathbb{E}(X \cdot g(Y)|Y) = \mathbb{E}(X|Y)g(Y)$

e) Independence: If X, Y are independent, then $\mathbb{E}(X|Y) = \mathbb{E}(X)$.

f) Functions of y : if $X = g(Y)$, then $\mathbb{E}(X|Y) = X$.

g) Jensen's inequality: if $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex, then $g(\mathbb{E}(X|Y)) \leq \mathbb{E}(g(X)|Y)$.

Remark.

If we take $h = 1$ in the definition of $\mathbb{E}(X|Y)$, we find that $\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$.

Corollary 1.1.7

Discrete case:

$$\begin{aligned}\mathbb{E}(X) &= \mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(\varphi(Y)) \text{ (where } \varphi(y) = \mathbb{E}(X|Y = y)) \\ &= \sum_y \varphi(y)\mathbb{P}(Y = y) = \sum_y \mathbb{E}(X|Y = y)\mathbb{P}(Y = y)\end{aligned}$$

i.e., the law of total probability.

Corollary 1.1.8

Continuous case:

$$\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(\varphi(Y)) = \int_{\mathbb{R}^m} \varphi(y)f_Y(y)dy = \int_{\mathbb{R}^m} \mathbb{E}(X|Y = y)f_Y(y)dy$$

Which again corresponds to the law of total probability.

Remark.

Conditional probabilities are a special case of conditional expectation; for an event G , $\mathbb{P}(G|Y) = \mathbb{E}(\mathbb{1}_G|Y)$ where $\mathbb{1}_G = 1$ if G occurs and 0 otherwise. (Note $\mathbb{P}(G) = \mathbb{E}(\mathbb{1}_G)$.)

1.2 Martingales

Definition 1.2.1: Martingale

Let (X_n) be a sequence of r.v.s (the observations). A sequence $(S_n, n \geq 1)$ of r.v.s is called a martingale (relative to the observations (X_n)) if, for all $n \geq 1$:

- a) $\mathbb{E}(|S_n|) < \infty$
- b) $\mathbb{E}(S_{n+1}|X_1, \dots, X_n) = S_n$

Remark.

If (S_n) is a martingale relative to (X_n) , then S_n is a function of $X_1, \dots, X_n$.
 ("We go through each of the X_i s once and remember what happened in the past.")
 Indeed, by b), $S_n = \psi_n(X_1, \dots, X_n) = \mathbb{E}(S_{n+1}|X_1 = x_1, \dots, X_n = x_n)$.

Remark.

Often, $S_n = X_n$.

The canonical interpretation: S_n is the wealth of a player at time n . At time n , the value of S_{n+1} is not yet known. The game is fair, because $\mathbb{E}(S_{n+1}|X_1, \dots, X_n) = S_n$.

Example 1.2.2

The "double or nothing" strategy.

Suppose we have a game where 1/2 of the time, the player receives twice his bet, and the rest of the time the player loses his bet.

If I bet x francs, my profit is x with probability 0.5 or $-x$ (also with probability 0.5).

Strategy: bet 1 CHF, then 2 CHF, ..., 2^n CHF until I win a bet, then immediately leave the game.

Then the total gain is $-1 - 2 - 4 - \dots - 2^{n-1} + 2^n = 1$.

So this strategy "guarantees" a profit of +1. But this is a bad strategy: let T be the number of games played up to the first win. Then:

$$\mathbb{P}(T = 1) = \frac{1}{2}, \mathbb{P}(T = 2) = \frac{1}{4}, \dots, \mathbb{P}(T = n) = \frac{1}{2^n}$$

$$\text{and } \mathbb{P}(T < \infty) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} (T = n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(T = n) = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

(Since the events are disjoint.)

Let L be the amount lost up to time $T - 1$. Then:

$$\mathbb{E}(L) = \sum_{n=1}^{\infty} \mathbb{E}(L|T = n) \mathbb{P}(T = n) = \sum_{n=2}^{\infty} (1 + 2 + \dots + 2^{n-2}) \frac{1}{2^n} = \sum_{n=2}^{\infty} \frac{2^{n-1} - 1}{2^n} = \infty$$

"Since most people only have finite amounts of wealth... this strategy tends to not work because people leave early when they go bankrupt."

Example 1.2.3

Let (X_n) be a series of i.i.d rv.s, with $\mathbb{E}(X_n) = 0$ (so they are centered and integrable). Define $S_n = X_1 + X_2 + \dots + X_n$.

Claim: Then (S_n) is a martingale.

Proof: We check the properties.

$$\text{i) } \mathbb{E}(|S_n|) = \mathbb{E}(|X_1 + \dots + X_n|) \leq \mathbb{E}(|X_1| + \dots + |X_n|) = n\mathbb{E}(|X_n|) < \infty \quad \checkmark$$

$$\text{ii) } \mathbb{E}(S_{n+1}|X_1, \dots, X_n) = \mathbb{E}(X_{n+1} + S_n|X_1, \dots, X_n)$$

$$\text{by linearity} = \mathbb{E}(X_{n+1}|X_1, \dots, X_n) + \mathbb{E}(S_n|X_1, \dots, X_n)$$

$$\text{by e) and f) } = \mathbb{E}(X_{n+1}) + S_n = 0 + S_n = S_n \quad \checkmark$$

Example 1.2.4

Now let (X_n) be i.i.d, $\mathbb{E}(X_1) = 0, \mathbb{E}(X_1^2) = \sigma^2 < \infty$. Define

$$Z_n = (\sum_{i=1}^n X_i)^2 - n\sigma^2 = S_n^2 - n\sigma^2.$$

Then (Z_n) is a martingale relative to (X_n) .

Proof. We check the properties.

$$\text{i) } \mathbb{E}(|Z_n|) \leq \mathbb{E}(S_n^2) + n\sigma^2 = \mathbb{E}((\sum_{i=1}^n X_i)^2) + n\sigma^2$$

$$\text{by independence} = \mathbb{E}\left(\sum_{i=1}^n X_i^2\right) + n\sigma^2 = n\sigma^2 + n\sigma^2 = 2n\sigma^2 < \infty.$$

$$\text{ii) } \mathbb{E}(Z_{n+1}|X_1, \dots, X_n) = \mathbb{E}((X_{n+1} + S_n)^2 - (n+1)\sigma^2|X_1, \dots, X_n)$$

$$= \mathbb{E}(X_{n+1}^2 + 2X_{n+1}S_n + S_n^2 - (n+1)\sigma^2|X_1, \dots, X_n)$$

$$\text{by a) } = \mathbb{E}(X_{n+1}^2|X_1, \dots, X_n) + 2\mathbb{E}(X_{n+1}S_n|X_1, \dots, X_n)$$

$$+ \mathbb{E}(S_n^2|X_1, \dots, X_n) - \mathbb{E}((n+1)\sigma^2|X_1, \dots, X_n)$$

$$\text{by e), d), f) } = \mathbb{E}(X_{n+1}^2) + 2S_n\mathbb{E}(X_{n+1}|X_1, \dots, X_n) + S_n^2 - (n+1)\sigma^2$$

$$\text{by e) } = \sigma^2 + 2S_n\mathbb{E}(X_{n+1}) + S_n^2 - (n+1)\sigma^2 = \sigma^2 + 2S_n(0) + S_n^2 - (n+1)\sigma^2$$

$$= \sigma^2 + S_n^2 - (n+1)\sigma^2 = S_n^2 - n\sigma^2 = Z_n$$

Example 1.2.5

"Likelihood ratio". Let (X_n) be i.i.d, and let f_0, f_1 be two bounded and positive density functions ($f_i > 0, \int_{\mathbb{R}} f_i(x)dx = 1$). Define:

$$R_n = \frac{f_1(X_1)f_1(X_2) \dots f_1(X_n)}{f_0(X_1)f_0(X_2) \dots f_0(X_n)}$$

Claim. If the common density of the X_n s is f_0 , then (R_n) is a martingale relative to (X_n) .

Proof. As always, the conditions must become checked. Let E_0 be the common density.

$$\begin{aligned} \text{i) } \mathbb{E}(|R_n|) &= E_0(R_n) = E_0 \left(\frac{f_1(X_1)f_1(X_2) \dots f_1(X_n)}{f_0(X_1)f_0(X_2) \dots f_0(X_n)} \right) \\ &= \int_{\mathbb{R}^n} \frac{f_1(X_1)f_1(X_2) \dots f_1(X_n)}{f_0(X_1)f_0(X_2) \dots f_0(X_n)} \cdot f_0(X_1)f_0(X_2) \dots f_0(X_n) dx_1 \dots dx_n = \\ &= \int_{\mathbb{R}^n} f_1(X_1)f_1(X_2) \dots f_1(X_n) \cdot dx_1 \dots dx_n = 1 < \infty \end{aligned}$$

$$\text{ii) } E_0(R_{n+1}|X_1 \dots X_n) = E_0 \left(R_n \frac{f_1(X_{n+1})}{f_0(X_{n+1})} | X_1 \dots X_n \right)$$

$$(\text{which by d)}) = R_n E_0 \left(\frac{f_1(X_{n+1})}{f_0(X_{n+1})} | X_1 \dots X_n \right)$$

$$\text{by e) } R_n E_0 \left(\frac{f_1(X_{n+1})}{f_0(X_{n+1})} \right) = R_n E_0 \frac{f_1(X_1)}{f_0(X_1)}$$

(Since the variables are i.i.d).

$$= R_n E_0(R_1) = R_n(1) \text{ (by a) for } n = 1) = R_n$$

Example 1.2.6

"Doob martingale". Let $X_1, X_2, \dots$ be arbitrary r.v.s, and let X be an iintegrable r.v., i.e. $\mathbb{E}(|X|) < \infty$. Define $M_n = \mathbb{E}(X|X_1 \dots X_n)$. Then (M_n) is a martingale relative to (X_n) .

Proof. First note that the map $x \rightarrow |x|$ is convex, so we will be able to use Jensen's inequality.

$$\text{i) } \mathbb{E}(|M_n|) = \mathbb{E}(|\mathbb{E}(X|X_1 \dots X_n)|) \leq \mathbb{E}(\mathbb{E}(|X||X_1 \dots X_n) = \mathbb{E}(|X|) < \infty.$$

(Since $\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$. Also used for ii).)

$$\text{ii) } \mathbb{E}(M_{n+1}|X_1 \dots X_n) = \mathbb{E}(\mathbb{E}(X|X_1 \dots X_n, X_{n+1})|X_1 \dots X_n)$$

$$\text{by c) } = \mathbb{E}(X|X_1 \dots X_n) = M_n.$$

Lemma 1.2.7

Let (S_n) be a martingale relative to (X_n) . Then:

$$\text{a) } \mathbb{E}(S_{n+m}|X_1 \dots X_n) = S_n \forall n, m \geq 1.$$

$$\text{b) } \mathbb{E}(S_n) = \mathbb{E}(S_1) \forall n \geq 1.$$

Proof. For a): $\mathbb{E}(S_{n+m}|X_1 \dots X_n)$ which by c) $= \mathbb{E}(\mathbb{E}(S_{n+m}|X_1 \dots X_n, \dots, X_{n+m-1})|X_1 \dots X_n)$
Which, by the definition of a martingale, $= \mathbb{E}(S_{n+m-1}|X_1 \dots X_n) = S_n$.

For b): $\mathbb{E}(S_n) = \mathbb{E}(\mathbb{E}(S_n|X_1))$ which by a) $= \mathbb{E}(S_1)$. □

1.3 Super- and submartingales

Definition 1.3.1: Supermartingale

(S_n) is a supermartingale relative to (X_n) if it fulfills three properties (where iii) was not necessary for regular martingales as we have it automatically):

$$\text{i) } \mathbb{E}(|S_n|) < \infty$$

$$\text{ii) } \mathbb{E}(S_{n+1}|X_1, \dots, X_n) \leq S_n$$

$$\text{iii) } S_n \text{ is a function of } X_1, \dots, X_n.$$

The intuition: S_n is above the value it would have if it were a true martingale.

Definition 1.3.2: Submartingale

In the same manner (S_n) is a supermartingale relative to (X_n) if it fulfills

- i) $\mathbb{E}(|S_n|) < \infty$
- ii) $\mathbb{E}(S_{n+1}|X_1, \dots, X_n) \geq S_n$
- iii) S_n is a function of $X_1, \dots, X_n$.

The intuition is the same: S_n is below the value it would have if it were a true martingale.

Remark.

1. For a true martingale, $S_n = \mathbb{E}(S_{n+1}|X_1, \dots, X_n)$, so S_n is a function of $X_1, \dots, X_n$.
2. If (S_n) is a supermartingale, then $(-S_n)$ is a submartingale (and vice versa).
3. S_n is a martingale iff (S_n) is a submartingale and a supermartingale.

Lemma 1.3.3

- a) Let (S_n) be a martingale relative to (X_n) . If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $\mathbb{E}(|\varphi(S_n)|) < \infty \forall n$, then $(\varphi(S_n), n \geq 1)$ is a submartingale.
- b) If (S_n) is a submartingale and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, nondecreasing, and $\mathbb{E}(|\varphi(S_n)|) < \infty \forall n$ as well, then $(\varphi(S_n), n \geq 1)$ is a submartingale.

Proof. For a). Conditions i) and iii) hold by the assumptions; we just need to check ii).

$$\mathbb{E}(\varphi(S_{n+1})|X_1 \dots X_n) \quad (\text{which by Jensen}) \geq \varphi(\mathbb{E}(S_{n+1}|X_1 \dots X_n)) = \varphi(S_n)$$

Which comes out from the definition of a martingale.

For b) we use Jensen's inequality again:

$$\mathbb{E}(\varphi(S_{n+1})|X_1 \dots X_n) \geq \varphi(\mathbb{E}(S_{n+1}|X_1 \dots X_n)) \geq \varphi(S_n)$$

□

Remark.

If (S_n) is a (super/sub)martingale, then for $n, m \geq 1$:

$$\mathbb{E}(S_{n+m}|X_1 \dots X_n) \leq / \geq S_n \text{ (respectively) and } \mathbb{E}(S_n) \leq / \geq \mathbb{E}(S_m)$$

1.4 Stopping

Motivation: A player decides to leave a sequence of games at a random time, that depends on the results of the previous games but NOT the future games.

Definition 1.4.1: Stopping time

Let (X_n) be an observation sequence (for ease, real-valued.) A stopping time T (relative to (X_n)) is an r.v with values in $\mathbb{N} \cup \{\infty\}$ s.t for all $k \in \mathbb{N}$, there is $B_k \subset \mathbb{R}^k$ such that $\{T = k\} = \{(X_1, \dots, X_k) \in B_k\}$.

For intuition, $\{T = k\}$ is the decision to stop at time k and B_k is a property of the observations $X_1, \dots, X_n$.

Example 1.4.2

Let (X_n) be i.i.d with $N(0, 1)$. At stage n , we observe X_n . Let T be the first time that $X_n \geq 2$. Then T is a stopping time.

Indeed: $k = 1 : \{T = 1\} = \{X_1 \geq 2\} = \{X_1 \in B_1\}$ where $B_1 = [2, \infty)$.

If $k \geq 2$, then $\{T = k\} = \{X_n < 2, \dots, X_{k-1} < 2, X_k \geq 2\} = \{(X_1, \dots, X_k) \in B_k\}$ where $B_k = (-\infty, 2) \times \dots (k-1 \text{ times}) \times (-\infty, 2) \times [2, \infty)$.

We don't usually write out B_k explicitly, but the concept can be useful.

Proposition 1.4.3

Some properties of stopping times.

- a) If T is a stopping time, then the sets $\{T \leq k\}$, $\{T < k\}$, $\{T > k\}$, and $\{T \geq k\}$ are determined by $(X_1, \dots, X_k)$.
- b) If $\{T \leq k\}$ is determined by $(X_1, \dots, X_k) \forall k$, then T is a stopping time.
- c) If S and T are two stopping times, then $S \wedge T = \min(S, T)$ and $\max(S, T)$ are also stopping times.
- d) If T is deterministic (chosen in advance, i.e. $k_0 \in \mathbb{N}$ such that $\mathbb{P}\{T = k_0\} = 1$), then T is a stopping time.
- e) If T is a stopping time, then for $k \geq 1$, $\mathbb{1}_{\{T=k\}}$ is a function of $(X_1, \dots, X_k)$.

Proof. For a): $\{T \leq k\} = \bigcup_{i=1}^k \{T = i\}$, which is determined by $(X_1, \dots, X_i)$ which eventually extends to X_k . The other equivalent properties come easily.

For b): Indeed, $\{T = k\} = \{T \leq k\} \cap \{T \leq k-1\}^C$, and we know that those components are determined by $(X_1, \dots, X_k)$.

For c): One side of the argument is $\{S \wedge T \leq k\} = \{S \leq k\} \cap \{T \leq k\}$ which are both appropriately determined. The idea is the same for the maximum.

For d): for $k \neq k_0$, $\{T = k\} = \{(X_1, \dots, X_k) \in \emptyset\}$ but for $k = k_0$, $\{T = k = k_0\} = \{(X_1, \dots, X_k) \in \mathbb{R}^{k_0}\}$.

For e): $\{T = k\} = \{(X_1, \dots, X_k) \in B_k\}$ for some $B_k \subset \mathbb{R}^k$. Then $\mathbb{1}_{\{T=k\}} = \mathbb{1}_{B_k}(X_1, \dots, X_k)$ where $\mathbb{1}_{B_k} : \mathbb{R}^k \rightarrow \mathbb{R}$ is governed by the rules: $\mathbb{1}_{B_k}(X_1, \dots, X_k) = 1$ if $(X_1, \dots, X_k) \in B_k$, and 0 otherwise. $\square$

Lemma 1.4.4

Let (S_n) be a (super/regular)martingale relative to (X_n) . Let T be a stopping time. For integers $n \geq k \geq 1$, $\mathbb{E}(S_n \mathbb{1}_{\{T=k\}}) = \mathbb{E}(S_k \mathbb{1}_{\{T=k\}})$.

Proof.

$$\mathbb{E}(S_n \mathbb{1}_{\{T=k\}}) = \mathbb{E}(\mathbb{E}(S_n \mathbb{1}_{\{T=k\}}) | X_1, \dots, X_k)$$

(as a consequence of conditional expectation definition. By d), and since $\mathbb{1}_{\{T=k\}}$ is a function of $X_1, \dots, X_k$ this equals:

$$= \mathbb{E}(\mathbb{1}_{\{T=k\}} \mathbb{E}(S_n | X_1, \dots, X_k)) \quad (\leq / =) \quad \mathbb{E}(\mathbb{1}_{\{T=k\}} S_k)$$

Since the right part of the first expression (super/regular) $(\leq / =) S_k$. $\square$

Lemma 1.4.5

Let W be a r.v such that $\mathbb{E}(|W|) < \infty$. Let T be a stopping time relative to (X_n) such that $\mathbb{P}\{T < \infty\} = 1$. Then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(W \mathbb{1}_{\{T > n\}}) = 0 \text{ and } \lim_{n \rightarrow \infty} \mathbb{E}(W \mathbb{1}_{\{T \leq n\}}) = \mathbb{E}(W)$$

Proof. We first establish these with W replaced by $|W|$.

$$(|W| \mathbb{1}_{\{T \leq n\}}) = \sum_{k=1}^n \mathbb{E}(|W| \mathbb{1}_{\{T=k\}}) = \sum_{k=1}^n \mathbb{E}(|W| | T = k) \mathbb{P}\{T = k\}$$

Since $|W| > 0$, we have a monotone increasing sequence, giving:

$$\sum_{k=1}^{\infty} \mathbb{E}(|W| | T = k) \mathbb{P}\{T = k\} = \mathbb{E}(|W|) < \infty$$

By the formula for total probability. Therefore:

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{E}(|W| \mathbf{1}_{\{T \leq n\}}) &= \mathbb{E}(|W|) \text{ and} \\ \lim_{n \rightarrow \infty} \mathbb{E}(|W| \mathbf{1}_{\{T > n\}}) &= \mathbb{E}(|W|(1 - \mathbf{1}_{\{T \leq n\}})) \\ &= \mathbb{E}(|W|) - \lim_{n \rightarrow \infty} \mathbb{E}(|W| \mathbf{1}_{\{T \leq n\}}) = \mathbb{E}(|W|) - \mathbb{E}(|W|) = 0.\end{aligned}$$

To remove the absolute values:

$$0 \leq |\mathbb{E}(W \mathbf{1}_{\{T > n\}})| \leq \mathbb{E}(W \mathbf{1}_{\{T > n\}}) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\implies \lim_{n \rightarrow \infty} \mathbb{E}(W \mathbf{1}_{\{T > n\}}) = 0$. Therefore: □

Question. Let (S_n) be a martingale relative to (X_n) . Then $\mathbb{E}(S_n) = \mathbb{E}(S_1), n \geq 1$. Suppose one player decides in advance to play n games, and the game is fair. Can a clever (honest) player do better.

i.e., the player decides to stop at a random time T , that is a stopping time. Is it possible to have $\mathbb{E}(S_T) > \mathbb{E}(S_1)$?

As an example, let X_n be i.i.d r.v.s, with $\mathbb{P}\{X_1 = 1\} = \frac{1}{2} = \mathbb{P}\{X_1 = -1\}$. Set $S_0, S_n = X_1 + \dots + X_n, n \geq 1$. Then (S_n) is a martingale.

Let T be the first time that we get to value 1, i.e. $T = \inf\{n \geq 1 : S_n = 1\}$. We will see later that $\mathbb{P}\{T < \infty\} = 1$ so $S_T = 1$ and indeed $\mathbb{E}(S_T) = 1 > 0 = \mathbb{E}(S_1)$. The problem with using this stopping time is that $\mathbb{E}(T) = \infty$.

Theorem 1.4.6: Optional stopping theorem

Let (S_n) be a martingale and T a stopping time relative to (X_n) . Suppose:

- a) $\mathbb{P}\{T < \infty\} = 1$
- b) $\mathbb{E}(|S_T|) < \infty$
- c) $\lim_{n \rightarrow \infty} \mathbb{E}(S_n | T > n) \mathbb{P}\{T > n\} = 0$.

Then $\mathbb{E}(S_T) = \mathbb{E}(S_1)$.

Remark.

- a) If $\mathbb{P}\{T = k_0\} = 1$ (i.e. the stopping time is a constant, decided random variable), then $\mathbb{E}(S_{k_0}) = \mathbb{E}(S_1)$, and the assumptions also hold.
- b) If $\mathbb{P}\{T = \infty\} > 0$, then $S_T = S_\infty$ and $\{T = \infty\}$ is not defined here.
- c) $\mathbb{E}(S_T)$ has a meaning by assumption b).

Lemma 1.4.7

Let T be a stopping time relative to (X_n) . Then $\mathbb{E}(S_{T \wedge n}) = \mathbb{E}(S_1)$.
 (The conclusion of the theorem is valid for bounded stopping times.)

Proof.

$$\mathbb{E}(S_1) = \text{(martingale)} = \mathbb{E}(S_n) = \sum_{k=1}^n \mathbb{E}(S_n \mathbf{1}_{\{T=k\}}) + \mathbb{E}(S_n \mathbf{1}_{\{T>n\}})$$

By Lemma 1,:

$$\begin{aligned} &= \sum_{k=1}^n \mathbb{E}(S_k \mathbf{1}_{\{T=k\}}) + \mathbb{E}(S_n \mathbf{1}_{\{T>n\}}) = \sum_{k=1}^n \mathbb{E}(S_T \mathbf{1}_{\{T=k\}}) + \mathbb{E}(S_n \mathbf{1}_{\{T>n\}}) \\ &= \mathbb{E}(S_T \mathbf{1}_{\{T \leq n\}}) + \mathbb{E}(S_n \mathbf{1}_{\{T>n\}}) = \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{T \leq n\}}) + \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{T>n\}}) = \mathbb{E}(S_{T \wedge n}) \end{aligned}$$

□

Now we prove the optional stopping theorem.

Proof. (Of the optional stopping theorem)

$$\begin{aligned} \mathbb{E}(S_T) &= \mathbb{E}(S_T \mathbf{1}_{\{T \leq n\}}) + \mathbb{E}(S_T \mathbf{1}_{\{T>n\}}) = \mathbb{E}(S_{T \wedge n} (1 - \mathbf{1}_{\{T>n\}})) + \mathbb{E}(S_T \mathbf{1}_{\{T>n\}}) \\ &= \mathbb{E}(S_{T \wedge n}) - \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{T>n\}}) + \mathbb{E}(S_T \mathbf{1}_{\{T>n\}}) \\ &= \mathbb{E}(S_1) - \mathbb{E}(S_n \mathbf{1}_{\{T>n\}}) + \mathbb{E}(S_T \mathbf{1}_{\{T>n\}}) \\ &= \mathbb{E}(S_1) - \mathbb{E}(S_n | T > n) \mathbb{P}\{T > n\} + \mathbb{E}(S_T \mathbf{1}_{\{T>n\}}) \rightarrow \mathbb{E}(S_1) \end{aligned}$$

Since by assumption c), as $n \rightarrow \infty$, the second term goes to 0, and also by lemma 2, the third term also goes to 0. □

Example 1.4.8

Returning to the example from earlier: random walks. Let X_n be i.i.d, and $\mathbb{P}\{X_1 = 1\} = \frac{1}{2} = \mathbb{P}\{X_1 = -1\}$. If $S_0 = 0, S_n = X_1 + \dots + X_n, n \geq 1$, then (S_n) is a martingale - the simple random walk on $\mathbb{Z}$.

Fix positive integers a and b . What is the probability that the r.w visits $-a$ before b ? Let $T = T_{-a,b} = \inf\{n \in \mathbb{N} : S_n = -a \text{ or } S_n = b\}$. First we see that T is a stopping time:

$$\{T = k\} = \{S_1 \notin \{-a, b\}, \dots, S_{k-1} \notin \{-a, b\}, S_k \in \{-a, b\}\}$$

We check the assumptions of the optional stopping theorem:

a) Need to show $\mathbb{P}\{T < \infty\} = 1$:

$$2^{-(a+b)} = \{X_1 = 1, X_2 = 1, \dots, X_{a+b} = 1\} \leq \mathbb{P}\{T \leq a + b\}$$

Where this inequality is true no matter where the r.w starts in the interval. Therefore the probability of not leaving $[-a, b]$ during $a + b$ units of time $= \mathbb{P}\{T > a + b\} = 1 - \mathbb{P}\{T \leq a + b\} = 1 - 2^{-(a+b)}$.

Then the probability of "not leaving $[-a, b]$ during $2(a + b)$ unites of time" = the probability of not leaving $[-a, b]$ during $a + b$ unites of time AND, starting from wherever we are at time $a + b$, not leaving $[-a, b]$ during the next $a + b$ units of time.

$$\begin{aligned} \mathbb{P}\{T > 2(a + b)\} &= \mathbb{E}(\mathbb{1}_{\{T > 2(a+b)\}}) = \mathbb{E}(\mathbb{1}_{\{T > a+b\}} \mathbb{1}_{\{T > 2(a+b)\}}) \\ &= \mathbb{E}(\mathbb{E}(\mathbb{1}_{\{T > a+b\}} \mathbb{1}_{\{T > 2(a+b)\}} | X_1, \dots, X_{a+b})) = \mathbb{E}(\mathbb{1}_{\{T > a+b\}} \mathbb{P}(T > 2(a + b) | X_1, \dots, X_{a+b})) \\ &\leq (1 - 2^{-(a+b)}) \mathbb{E}(\mathbb{1}_{\{T > a+b\}}) = (1 - 2^{-(a+b)}) \mathbb{P}\{T > a + b\} = (1 - 2^{-(a+b)})^2 \\ &\quad \vdots \\ \mathbb{P}\{T > k(a + b)\} &\leq (1 - 2^{-(a+b)})^k < 1 \end{aligned}$$

So as $k \rightarrow \infty, 0 \leq \mathbb{P}\{T = \infty\} \leq 0 \implies \mathbb{P}\{T = \infty\} = 0$.

b) We need to show $\mathbb{E}(|S_T|) < \infty$:

$$\mathbb{E}(|S_T|) \leq \max(a, b) < \infty \quad (\text{since } |S_T| \leq \max(a, b))$$

c) We need to show $\lim_{n \rightarrow \infty} |\mathbb{E}(S_n | T > n) \mathbb{P}\{T > n\}| = 0$.

$$= \lim_{n \rightarrow \infty} |\mathbb{E}(S_n \mathbb{1}_{\{T > n\}})| \leq \lim_{n \rightarrow \infty} |\mathbb{E}(|S_n| \mathbb{1}_{\{T > n\}})|$$

Since (for the last step) $\mathbb{P}\{T < \infty\} = 1$:

$$\leq \lim_{n \rightarrow \infty} \max(a, b) \mathbb{P}\{T > n\} = 0$$

Example 1.4.9

Random walks continued.

So we have all the conditions. Then as a conclusion of the optional stopping theorem:

$$\mathbb{E}(S_T) = \mathbb{E}(S_1) = \mathbb{E}(X_1) = 0 = -a\mathbb{P}\{S_T = -a\} + b\mathbb{P}\{S_T = b\}.$$

We return to the original question: $-a$ or b first? Let $P_{-a,b} = \mathbb{P}\{S_T = -a\} = 1 - \mathbb{P}\{S_T = b\}$.

Then $-aP_{-a,b} + b(1 - P_{-a,b}) = 0$ so $P_{-a,b} = \frac{b}{a+b}$ = probability that the r.w visits $-a$ before b , and if we want the converse (b before $-a$), it's $\frac{a}{a+b}$.

Now let's consider just the first visit to b , and not consider a . Let $T_b = \inf\{n \geq 0 : S_n = b\}$ ($b > 0$). Notice that $\{S_{T_{-a,b}} = b\} = \{T_b \leq T_{-a}, T_b < \infty\}$ and

$$\{T_b < \infty\} \subset \left\{ \min_{n \in \{0, \dots, T_b\}} S_n > -\infty \right\} \subset \bigcup_{a \in \mathbb{N}} \{T_b = T_{\{-a, b\}}\}$$

In particular:

$$\mathbb{P}\{T_b < \infty\} = \mathbb{P}\left(\bigcup_{a \in \mathbb{N}} (\{T_b = T_{-a,b}\} \cap \{T_b < \infty\})\right) = \mathbb{P}\left(\bigcup_{a \in \mathbb{N}} \{T_b \leq T_{-a}, T_b < \infty\}\right)$$

$$\lim_{a \rightarrow \infty} \mathbb{P}\{T_b \leq T_{-a}, T_b < \infty\} = \lim_{a \rightarrow \infty} \mathbb{P}\{S_{T_{-a,b}} = b\} = \lim_{a \rightarrow \infty} \frac{a}{a+b} = 1.$$

"There's never a level that we do not visit at all".

Example 1.4.10

Now we want to work on the mean time to visit b , i.e. we want to determine $\mathbb{E}(T_b)$. Define $Y_n = S_n^2 - n$. As seen earlier (first lecture?), (Y_n) is a martingale. We want to apply the optional stopping theorem to (Y_n) and $T_{-a,b}$. We check the hypotheses:

a) $\mathbb{P}\{T < \infty\} = 1$: already done.

b) $\mathbb{E}(|Y_T|) < \infty$:

$$\mathbb{E}(|Y_T|) = \mathbb{E}(|S_T^2 - T|) \leq \mathbb{E}(S_T^2) + \mathbb{E}(T) \leq \max(a^2, b^2) + \mathbb{E}(T).$$

So we need $\mathbb{E}(T) = \mathbb{E}(T_{-a,b})$ to be finite. Recall from probability that if $X \geq 0$ is a r.v, then $\mathbb{E}(X) \leq 1 + \sum_{k=1}^{\infty} \mathbb{P}\{X \geq k\}$. Note also: if $\mathbb{P}\{X \in \mathbb{N}\} = 1$, then $\mathbb{E}(X) = \sum_{k=1}^{\infty} \mathbb{P}\{X \geq k\}$.

We will use the first part since this is not integer-valued. Therefore:

$$\begin{aligned} \mathbb{E}\left(\frac{T}{a+b}\right) &\leq 1 + \sum_{k=1}^{\infty} \mathbb{P}\left\{\frac{T}{a+b} \geq k\right\} \leq 1 + \sum_{k=1}^{\infty} \mathbb{P}\left\{\frac{T}{a+b} > k-1\right\} \\ &\leq 1 + \sum_{k=1}^{\infty} (1 - 2^{-(a+b)})^{k-1} \text{ (from last time); since } (1 - 2^{-(a+b)}) < 1, < \infty. \end{aligned}$$

So $\mathbb{E}(T) < \infty$, which establishes b).

c) $\mathbb{E}(Y_n \mathbf{1}_{\{T > n\}}) = \mathbb{E}((S_n^2 - n) \mathbf{1}_{\{T > n\}}) = \mathbb{E}(S_n^2 \mathbf{1}_{\{T > n\}}) - \mathbb{E}(n \mathbf{1}_{\{T > n\}})$.

Now $\mathbb{E}(S_n^2 \mathbf{1}_{\{T > n\}}) \leq \max(a^2, b^2) \mathbb{P}\{T > n\} \rightarrow 0$ as $n \rightarrow \infty$ since $\mathbb{P}\{T < \infty\} = 1$.

Also, $\mathbb{E}(n \mathbf{1}_{\{T > n\}}) \leq \mathbb{E}(T \mathbf{1}_{\{T > n\}}) \rightarrow 0$ as $n \rightarrow \infty$ as well (by a former lemma (1.4.5)? with $W = T$, since $\mathbb{E}(T) < \infty$.)

So we have checked all the assumptions. By the optional stopping theorem:

$$\mathbb{E}(Y_T) = \mathbb{E}(Y_1) = \mathbb{E}(S_1^2 - 1) = \mathbb{E}(X_1^2 - 1) = \mathbb{E}(1 - 1) = 0.$$

$0 = \mathbb{E}(S_T^2 - T)$, therefore:

$$\begin{aligned} \mathbb{E}(T) &= \mathbb{E}(S_T^2) = a^2 \mathbb{P}\{S_{T_{-a,b}} = -a\} + b^2 \mathbb{P}\{S_{T_{-a,b}} = b\} \\ &= a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = ab \left(\frac{a}{a+b} + \frac{b}{a+b} \right) = ab. \end{aligned}$$

$\implies \mathbb{E}(T_{-a,b}) = ab \implies \mathbb{E}(T_{-a,a}) = a^2$, the meaning time for moving a steps away from the starting point.

Now we calculate $\mathbb{E}(T_b)$. For all $a \in \mathbb{N}^+$, $T_b \geq T_{-a,b}$ because if $-a$ is visited before b , $T_{-a,b} < T_b$ and if b is visited before $-a$, $T_{-a,b} = T_b$. Therefore $\mathbb{E}(T_b) \geq \mathbb{E}(T_{-a,b}) = ab$, $\forall a \in \mathbb{N}$. In particular $\mathbb{E}(T_b) \geq \sup_{a \in \mathbb{N}}(ab)$ which since $b \geq 1 = \infty$. In particular, $\mathbb{E}(T_1) = \infty$.

A student: "You mean, we are starting at 0, and we would just have to go up one to get to 1, it would still take on average infinitely long. Isn't that counter-intuitive?"

"Yes." - Dalang

Theorem 1.4.11: Doob decomposition theorem

Let (S_n) be a supermartingale relative to (X_n) . There exists a martingale (M_n) and a process (A_n) such that $n \rightarrow A_n$ is nondecreasing, $A_0 = 0$, A_{n+1} is a function of $(X_1, \dots, X_n)$ [i.e. (A_n) is predictable] and $S_n = M_n - A_n$.

Moreover, this decomposition is unique.

Proof. See Problem set 4. □

Lemma 1.4.12

Let (S_n) be a supermartingale and T a stopping time relative to (X_n) . Then $\mathbb{E}(S_{T \wedge n}) \leq \mathbb{E}(S_1)$.

This is similar to the lemma from earlier for martingales, but, unlike most of the time, here you can't just go through and replace the equal signs with less/greater than signs.

Proof. Let $S_n = M_n - A_n$ be the Doob decomposition of (S_n) . Then:

$$S_{T \wedge n} = M_{T \wedge n} - A_{T \wedge n} \text{ so } \mathbb{E}(S_{T \wedge n}) = \mathbb{E}(M_{T \wedge n}) - \mathbb{E}(A_{T \wedge n})$$

Since $A_{T \wedge n} \geq 0$:

$$\leq \mathbb{E}(M_{T \wedge n}) = \mathbb{E}(M_1) = \mathbb{E}(S_1)$$

By first the lemma (3) from earlier for martingales, and the fact that $A_1 = 0$. □

Theorem 1.4.13: Optional stopping theorem for supermartingales

Let (S_n) be a supermartingale and T a stopping time relative to (X_n) . Suppose that $\mathbb{P}\{T < \infty\} = 1$ and there is an r.v $W \geq 0$ with $\mathbb{E}(W) < \infty$ and $S_{T \wedge n} \geq -W, \forall n \in \mathbb{N}$. Then $\mathbb{E}(S_T) \leq \mathbb{E}(S_1)$.

Remark.

We often use this when (S_n) is nonnegative (take $W = 0$).

Proof. Fix $b > 0$, and let $S_n^b = \min(b, S_n)$, $n \geq 1$. Then (S_n^b) is also a supermartingale because $x \rightarrow \min(b, x)$ is concave and nondecreasing (we saw the corresponding statement for submartingales and convex nondecreasing functions).

(*) By Lemma 4, $\mathbb{E}(S_1^b) \geq \mathbb{E}(S_{T \wedge n}^b)$, $n \geq 1$. Then:

$$\begin{aligned} 0 &\leq |\mathbb{E}(S_{T \wedge n}^b) - \mathbb{E}(S_T^b)| = |\mathbb{E}(S_{T \wedge n}^b - S_T^b)| = |\mathbb{E}((S_n^b - S_T^b)\mathbf{1}_{\{T > n\}})| \\ &\leq \mathbb{E}((b + W)\mathbf{1}_{\{T > n\}}) \rightarrow 0 \text{ as } n \rightarrow \infty \end{aligned}$$

(By Lemma 2, and since $\mathbb{E}(b + W) < \infty$).

Therefore $\mathbb{E}(S_{T \wedge n}^b) \rightarrow \mathbb{E}(S_T^b)$ as $n \rightarrow \infty$. Now let $n \rightarrow \infty$ at (*). Then $\mathbb{E}(S_1^b) \geq \mathbb{E}(S_T^b)$ (**). As $b \uparrow \infty$: $S_n^b \uparrow S_n$, $S_T^b \uparrow S_T$. By the monotone convergence theorem:

$$\lim_{b \rightarrow \infty} \mathbb{E}(S_T^b) = \mathbb{E}(S_T) \implies \mathbb{E}(S_1) \geq \mathbb{E}(S_T)$$

By referencing (**). □

Proposition 1.4.14

Let (S_n) be a supermartingale and T a stopping time. Suppose that $\mathbb{P}\{S_n \geq 0\} = 1$. Then:

$$\mathbb{E}(S_1) \geq \mathbb{E}(S_T \mathbf{1}_{\{T < \infty\}})$$

Proof. By lemma 4 and since $S_n \geq 0$:

$$\mathbb{E}(S_1) \geq \mathbb{E}(S_{T \wedge n}) \geq \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{T \leq n\}}) = \mathbb{E}(S_T \mathbf{1}_{\{T \leq n\}})$$

On $\{T < \infty\}$, $S_T \mathbf{1}_{\{T \leq n\}} = 1$ for $n \geq T$ and 0 otherwise. Then $n \rightarrow S_T \mathbf{1}_{\{T \leq n\}}$ is nondecreasing and it converges to $S_T = S_T \mathbf{1}_{\{T < \infty\}}$.

So on $\{T < \infty\}$, $S_T \mathbf{1}_{\{T < \infty\}} = S_T$.

On $\{T = \infty\}$, $S_T \mathbf{1}_{\{T \leq n\}} = 0 = S_T \mathbf{1}_{\{T < \infty\}}$. In both cases, $S_T \mathbf{1}_{\{T \leq n\}} \uparrow S_T \mathbf{1}_{\{T < \infty\}}$. By the monotone convergence theorem:

$$\lim_{n \rightarrow \infty} \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{T \leq n\}}) = \mathbb{E}(S_T \mathbf{1}_{\{T < \infty\}}) \leq \mathbb{E}(S_1)$$

□

Theorem 1.4.15

Let (S_n) be a supermartingale and U and T be two stopping times relative to (X_n) . Suppose there is $N \in \mathbb{N}$ such that $\mathbb{P}\{1 \leq U \leq T \leq N\} = 1$. Then $\mathbb{E}(S_T) \leq \mathbb{E}(S_U)$.

Proof. 1) Fix $k < N$. For $k \leq n \leq N$:

$$\begin{aligned} \mathbb{E}(S_{n+1} \mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}}) &= \mathbb{E}(\mathbb{E}(S_{n+1} \mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}} | X_1, \dots, X_n)) \\ &= \mathbb{E}(\mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}} \mathbb{E}(S_{n+1} | X_1, \dots, X_n)) \leq \mathbb{E}(\mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}} S_n) \end{aligned}$$

2) $n \rightarrow \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{U=k\}})$ is nondecreasing on $[k, N]$. Indeed:

$$\begin{aligned} \mathbb{E}(S_{T \wedge n} \mathbf{1}_{\{U=k\}}) &= \mathbb{E}(S_T \mathbf{1}_{\{T \leq n\}} \mathbf{1}_{\{U=k\}}) + \mathbb{E}(S_T \mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}}) \\ &\geq \mathbb{E}(S_T \mathbf{1}_{\{T \leq n\}} \mathbf{1}_{\{U=k\}}) + \mathbb{E}(S_{n+1} \mathbf{1}_{\{T > n\}} \mathbf{1}_{\{U=k\}}) = \mathbb{E}(S_{T \wedge (n+1)} \mathbf{1}_{\{U=k\}}) \end{aligned}$$

By 2), $\mathbb{E}(S_{T \wedge k} \mathbf{1}_{\{U=k\}}) \geq \mathbb{E}(S_{T \wedge N} \mathbf{1}_{\{U=k\}})$, so $\mathbb{E}(S_k \mathbf{1}_{\{U=k\}}) \geq \mathbb{E}(S_T \mathbf{1}_{\{U=k\}})$ because $T \geq U = k$.

3) Finally, $\mathbb{E}(S_U) = \sum_{k=1}^N \mathbb{E}(S_k \mathbf{1}_{\{U=k\}}) \geq \sum_{k=1}^N \mathbb{E}(S_T \mathbf{1}_{\{U=k\}}) = \mathbb{E}(S_T \sum_{k=1}^N \mathbf{1}_{\{U=k\}}) = \mathbb{E}(S_T)$. $\square$

Chapter 2: Martingale theory

2.1 Martingale convergence theorems

Recall the different ways of convergence of a sequence of r.v.s (S_n) to an r.v S .

1) Convergence in probability: $S_n \rightarrow S$ in probability if:

$$\forall \varepsilon > 0, \lim_{n \rightarrow \infty} \mathbb{P}\{|S_n - S| > \varepsilon\} = 0.$$

2) Almost sure convergence: $S_n \rightarrow S$ a.s. if:

$$\mathbb{P}\{w \in \Omega : \lim_{n \rightarrow \infty} S_n(w) = S(w)\} = 1.$$

3) Monotone convergence theorem: If $\mathbb{P}\{S_n \geq 0\} = 1, \mathbb{P}\{S_n \leq S_{n+1}\} = 1, \forall n \in \mathbb{N}$, and $S_n \rightarrow S$ a.s, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(S_n) = \mathbb{E}(\lim_{n \rightarrow \infty} S_n) = \mathbb{E}(S).$$

4) Dominated convergence theorem: If $S_n \rightarrow S$ a.s, and if there is an r.v $W \geq 0$ such that $\mathbb{E}(W) < \infty$, and $\mathbb{P}\{|S_n| \leq W\} = 1, \forall n \in \mathbb{N}$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(S_n) = \mathbb{E}(\lim_{n \rightarrow \infty} S_n) = \mathbb{E}(S).$$

Proposition 2.1.1

Let (X_n) be a sequence of r.v.s, X an r.v. Define:

$$A_n(\varepsilon) = \{|X_n - X| > \varepsilon\}, B_m(\varepsilon) = \bigcup_{n:n \geq m} A_n(\varepsilon)$$

Then:

- a) $X_n \rightarrow X$ a.s iff $\forall \varepsilon > 0, \lim_{m \rightarrow \infty} \mathbb{P}(B_m(\varepsilon)) = 0$ iff $\forall \varepsilon > 0, \lim_{m \rightarrow \infty} \mathbb{P}\{\sup_{n:n \geq m} |X_n - X| > \varepsilon\} = 0$.
- b) If $\sum_{n=1}^{\infty} \mathbb{P}(A_n(\varepsilon)) < \infty$, then $X_n \rightarrow X$ a.s.
- c) If $X_n \rightarrow X$ a.s, then $X_n \rightarrow X$ in probability as well.

Proof. a) Observe that:

$$\left\{ \lim_{n \rightarrow \infty} X_n = X \right\} = \bigcap_{k \in \mathbb{N}^*} \bigcup_{N \in \mathbb{N}} \bigcap_{n:n \geq N} \left\{ |X_n - X| \leq \frac{1}{k} \right\}$$

because: $\lim_{n \rightarrow \infty} X_n(w) = X(w)$ iff $\forall k \in \mathbb{N}^*, \exists N \in \mathbb{N}$ such that $n \geq N \implies |X_n(w) - X(w)| \leq \frac{1}{k}$

Therefore:

$$\begin{aligned} \mathbb{P} \left\{ \lim_{n \rightarrow \infty} X_n = X \right\} = 1 &\iff \mathbb{P} \left(\left\{ \lim_{n \rightarrow \infty} X_n = X \right\}^c \right) = 0 \\ &\iff \mathbb{P} \left(\bigcup_{k \in \mathbb{N}^*} \bigcap_{N \in \mathbb{N}} \bigcup_{n:n \geq N} \left\{ |X_n - X| > \frac{1}{k} \right\} \right) = \mathbb{P} \left(\bigcup_{k \in \mathbb{N}^*} \bigcap_{N \in \mathbb{N}} B_N \left(\frac{1}{k} \right) \right) = \mathbb{P} \left(\bigcup_{k \in \mathbb{N}^*} C_k \right) \end{aligned}$$

(Intersections become unions, inequality flips etc, due to complement.)

Where we let $C_k = \bigcap_{N \in \mathbb{N}} B_N \left(\frac{1}{k} \right)$, and note that $C_k \subset C_{k+1}$. The above says that $\mathbb{P}(\bigcup_{k \in \mathbb{N}^*} C_k) = 0 \iff \forall k \in \mathbb{N}^*, \mathbb{P}(C_k) = 0$.

Sp $0 = \mathbb{P}(C_k) = \mathbb{P}(\bigcap_{N \in \mathbb{N}} B_N \left(\frac{1}{k} \right)) = \lim_{N \rightarrow \infty} \mathbb{P}(B_N \left(\frac{1}{k} \right))$.

Since $B_N \left(\frac{1}{k} \right) \supset B_{N+1} \left(\frac{1}{k} \right)$, it follows that:

$$(*) \quad \mathbb{P} \left\{ \lim_{n \rightarrow \infty} X_n = X \right\} = 1 \iff \forall k \in \mathbb{N}^*, \lim_{N \rightarrow \infty} \mathbb{P}(B_N \left(\frac{1}{k} \right)) = 0$$

It remains to check that this is equivalent to: $(**)$ $\forall \varepsilon > 0, \lim_{n \rightarrow \infty} \mathbb{P}(B_N(\varepsilon)) = 0$. Clearly $(**) \implies (*)$, but we need to check the opposite implication:

For $\varepsilon < \varepsilon', A_n(\varepsilon') \subset A_n(\varepsilon)$ so $B_n(\varepsilon) \supset B_n(\varepsilon')$. If $\lim_{N \rightarrow \infty} \mathbb{P}\{B_N(\varepsilon)\} = 0$, it implies $\lim_{N \rightarrow \infty} \mathbb{P}\{B_N(\varepsilon')\} = 0$. This shows $(*) \implies (**)$ and we are done with a). $\square$

Proof. b) $\mathbb{P}(B_N(\varepsilon)) \leq \sum_{n=N}^{\infty} \mathbb{P}(A_n(\varepsilon))$, so if $\sum_{n=1}^{\infty} \mathbb{P}(A_n(\varepsilon)) < \infty$, then:

$$0 \leq \lim_{N \rightarrow \infty} \mathbb{P}(B_N(\varepsilon)) \leq \lim_{N \rightarrow \infty} \sum_{n=N}^{\infty} \mathbb{P}(A_n(\varepsilon)) = 0.$$

c) $X_n \rightarrow X$ in probability iff. $\forall \varepsilon > 0, \lim_{n \rightarrow \infty} \mathbb{P}(A_n(\varepsilon)) = 0$. If $X_n \rightarrow X$ a.s, then:

$$0 \leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n(\varepsilon)) \leq \lim_{n \rightarrow \infty} (B_n(\varepsilon)) = 0.$$

□

Lemma 2.1.2: Cauchy criterion for almost-sure convergence

Let (S_n) be a sequence of r.vs such that:

$$\forall \varepsilon > 0, \lim_{m \rightarrow \infty} \mathbb{P} \left\{ \sup_{i:i>m} |S_i - S_m| > \varepsilon \right\} = 0$$

Then there exists an r.v S such that $S_n \rightarrow S$ a.s.

Proof: see problem set 5.

Theorem 2.1.3: Martingale convergence theorem

Let (S_n) be a martingale relative to (X_n) . Suppose that there is $M < \infty$ such that $\mathbb{E}(S_n^2) \leq M, \forall n \in \mathbb{N}$, (L^2 -bounded martingale).

Then there is an r.v S such that $\lim_{n \rightarrow \infty} S_n = S$ a.s.

Lemma 2.1.4: The Doob-Kolmogorov inequality

Let (S_n) be a martingale relative to (X_n) such that $\mathbb{P}\{S_n > 0\} = 1 \forall n \in \mathbb{N}$. Then for all $\lambda > 0$:

$$\mathbb{P} \left\{ \max_{i=1, \dots, n} S_i \geq \lambda \right\} \leq \frac{\mathbb{E}(S_n)}{\lambda}$$

Proof. Set $T = \inf\{k \geq 0 : S_k \geq \lambda\}$ if $\max_{i=1, \dots, n} S_i \geq \lambda$ and $T = n$ if $\max_{i=1, \dots, n} S_i < \lambda$.

Then T is a stopping time relative to (X_n) : for $k < n$, $\{T = k\} = \{S_1 < \lambda, \dots, S_{k-1} < \lambda, S_k \geq \lambda\}$, where all of the S_j s are determined by $X_1, \dots, X_k$.

For $k = n$: $\{T = n\} = \{S_1 < \lambda, \dots, S_{n-1} < \lambda, S_n < \lambda\}$, where all of the S_n s are determined by $X_1, \dots, X_n$.

By the optional stopping theorem for bounded stopping times:

$$\mathbb{E}(S_n) \geq \mathbb{E}(S_T) \geq \mathbb{E}(S_T \mathbb{1}_{\{\max_{i=1, \dots, n} S_i \geq \lambda\}})$$

$$\geq \mathbb{E}(\lambda \mathbf{1}_{\{\max_{i=1,\dots,n} S_i \geq \lambda\}}) = \lambda \mathbb{P}\left\{\max_{i=1,\dots,n} S_i \geq \lambda\right\}$$

□

Lemma 2.1.5: Variant on this inequality

Fix $p \geq 1$. If $\mathbb{E}(|S_i|^p) < \infty, i = 1, \dots, n$, then:

$$\mathbb{E}\left(\max_{i=1,\dots,n} |S_i|^p\right) \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}(|S_n|^p)$$

For a proof, see problem set 6. As a special case, when $p = 2$, $\mathbb{E}(\max_{i=1,\dots,n} S_i^2) \leq 4\mathbb{E}(S_n^2)$.

Proof. Of the martingale convergence theorem. Observe that $n \rightarrow \mathbb{E}(S_n^2)$ is nondecreasing, because (S_n^2) is a submartingale (squaring gives a submartingale.) In addition:

$$\mathbb{E}(S_{n+m}^2) = \mathbb{E}((S_m + S_{n+m} - S_m)^2) = \mathbb{E}(S_m^2) + 2\mathbb{E}(S_m(S_{n+m} - S_m)) + \mathbb{E}((S_{n+m} - S_m)^2) \quad (*)$$

Treating just the second term:

$$\begin{aligned} \mathbb{E}(S_m(S_{n+m} - S_m)) &= \mathbb{E}(\mathbb{E}(S_m(S_{n+m} - S_m) | X_1, \dots, X_m)) \\ &= \mathbb{E}(S_m \mathbb{E}(S_{n+m} - S_m | X_1, \dots, X_n)) = S_m - S_m = 0 \end{aligned}$$

(By the martingale property of S_m .) So we get from (*) that:

$$\mathbb{E}(S_{n+m}^2) = \mathbb{E}(S_m^2) + \mathbb{E}((S_{n+m} - S_m)^2)$$

Since $m \rightarrow \mathbb{E}(S_m^2)$ is nondecreasing and bounded by M , we can set $M_0 = \lim_{m \rightarrow \infty} \mathbb{E}(S_m^2)$; $M_0 \leq M$.

For $m \in \mathbb{N}$, define $Y_n = S_{m+n} - S_m$. Then (Y_n) is a martingale relative to (Y_n) , indeed:

$$\begin{aligned} \mathbb{E}(Y_{n+1} | Y_1, \dots, Y_n) &= \mathbb{E}(\mathbb{E}(Y_{n+1} | X_1, \dots, X_{n+m}) | Y_1, \dots, Y_n) \\ &= \mathbb{E}(\mathbb{E}(S_{m+n+1} - S_m | X_1, \dots, X_{n+m}) | Y_1, \dots, Y_n) = \mathbb{E}(S_{m+n} - S_m | Y_1, \dots, Y_n) \end{aligned}$$

[Again by the martingale property of (S_n)]

$$= \mathbb{E}(Y_n | Y_1, \dots, Y_n) = Y_n.$$

Now we apply the Doob-Kolmogorov inequality to the submartingale (Y_n^2) :

$$\mathbb{P} \left\{ \max_{i=1, \dots, n} |Y_i| > \varepsilon \right\} = \mathbb{P} \left\{ \max_{i=1, \dots, n} Y_i^2 > \varepsilon^2 \right\} \leq \frac{1}{\varepsilon^2} \mathbb{E}(Y_n^2)$$

That is: $\mathbb{P} \left\{ \max_{i=1, \dots, n} |S_{m+i} - S_m| > \varepsilon \right\} \leq \frac{1}{\varepsilon^2} \mathbb{E}((S_{m+n} - S_m)^2)$. Next:

$$\mathbb{P} \left\{ \max_{i=m, \dots, m+n} |S_i - S_m| > \varepsilon \right\} \leq \frac{1}{\varepsilon^2} (\mathbb{E}(S_{m+n}^2) - \mathbb{E}(S_m^2))$$

We want to let $n \rightarrow \infty$:

$$\bigcup_{n=1}^{\infty} \left\{ \max_{i=m, \dots, m+n} |S_i - S_m| > \varepsilon \right\} = \left\{ \sup_{i \geq m} |S_i - S_m| > \varepsilon \right\}$$

Therefore $P \left\{ \sup_{i \geq m} |S_i - S_m| > \varepsilon \right\} \leq \frac{1}{\varepsilon^2} (M_0 - \mathbb{E}(S_m^2))$.

Let $m \rightarrow \infty$: then $0 \leq \lim_{m \rightarrow \infty} P \left\{ \sup_{i \geq m} |S_i - S_m| > \varepsilon \right\} \leq \frac{1}{\varepsilon^2} (M_0 - M_0) = 0$. $\square$

Definition 2.1.6: Upcrossing

Let (S_n) be a submartingale. For $a < b, N \in \mathbb{N}^+$, let $V_{a,b,N}$ be the number of upcrossings of $[a, b]$ between times 1 and N , i.e. the number of couples (i, j) with $1 \leq i < j \leq N$ such that $S_i \leq a, a < S_k < b$ for $i < k < j$, and $S_j \geq b$.

"It starts below the interval $[a, b]$ at time i , moves around for a while in the interval, and exits above the interval by time j ."

Remark.

If $f : \mathbb{N} \rightarrow \mathbb{R}$ is deterministically nondecreasing, there exists at most one upcrossing of $[a, b]$ during $\{1, \dots, N\}$.

Lemma 2.1.7: Upcrossings inequality

Let (S_n) be a submartingale and $V_{a,b,N}$ be as above. Then:

$$\mathbb{E}(V_{a,b,N}) \leq \frac{1}{b-a} (\mathbb{E}((S_N - a)_+) - \mathbb{E}((S_1 - a)_+))$$

Where $x_+ = \max(x, 0)$.

Proof. Set $X_n = (S_n - a)_+$. Since $x \rightarrow (x - a)_+$ is nondecreasing and convex, (X_n) is also

a submartingale. Set $T_0 = 0$, and for $k = 1, \dots, N$, if k is odd:

$$T_k = \begin{cases} N & \text{if } X_j > 0 \text{ for } j \in [T_{k-1} + 1] \\ \min\{j \in [T_{k-1} + 1, N], X_j = 0\} & \text{otherwise} \end{cases}$$

If k is even, then:

$$T_k = \begin{cases} N & \text{if } X_j < b - a \text{ for } j \in [T_{k-1} + 1] \\ \min\{j \in [T_{k-1} + 1, N], X_j \geq b - a\} & \text{otherwise} \end{cases}$$

The T_k are stopping times $1 \leq T_k \leq T_{k-1} \leq N$, so they are the right-endpoints of an upcrossing or a downcrossing.

If for some $k \leq N$, $T_k = N$, then $T_{k+1} = N$. By the optional stopping theorem for bounded stopping times, $\mathbb{E}(X_{T_k}) \leq \mathbb{E}(X_{T_{k+1}})$. This looks strange for even k , but it works out ok. Notice that:

$$X_N - X_1 = \sum_{k=1}^N (X_{T_{k+1}} - X_{T_k}) + X_{T_1} - X_1 = \sum_{k \text{ odd} \in \{1, \dots, N\}} (X_{T_{k+1}} - X_{T_k}) + \sum_{k \text{ even} \in \{1, \dots, N\}} (X_{T_{k+1}} - X_{T_k}) + X_{T_1} - X_1$$

Since the first term $\geq (b - a)V_{a,b,N}$, taking expectations:

$$\begin{aligned} \mathbb{E}(X_N - X_1) &\geq (b - a)\mathbb{E}(V_{a,b,N}) + \sum_{k \text{ even} \in \{1, \dots, N\}} (\mathbb{E}(X_{T_{k+1}}) - \mathbb{E}(X_{T_k})) + \mathbb{E}(X_{T_1}) - \mathbb{E}(X_1) \\ &\geq (b - a)\mathbb{E}(V_{a,b,N}) \end{aligned}$$

□

Theorem 2.1.8: A.s convergence of submartingales

Let (S_n) be a submartingale relative to (X_n) , such that $\sup_{n \in \mathbb{N}} \mathbb{E}(S_n^+) < \infty$, where $x^+ = \max(x, 0)$.

Then there is an r.v S such that $S_n \rightarrow S$ a.s and $\mathbb{E}(|S|) < \infty$.

Proof. Fix $a < b$, $N \in \mathbb{N}$ and $V_{a,b,N}$ as before. Then:

$$\mathbb{E}(V_{a,b,N}) \leq \frac{1}{b-a} \mathbb{E}((S_N - a)_+) \leq \frac{1}{b-a} (\mathbb{E}(S_N^+) + |a|) \leq \frac{1}{b-a} (\sup_{N \in \mathbb{N}} \mathbb{E}(S_N^+) + |a|) < \infty$$

Since $(x - a)_+ \leq x_+ + |a| \forall x, a \in \mathbb{R}$. Then for $N \uparrow \infty$, by definition, $V_{a,b,N} \uparrow V_{a,b} = \#$ of upcrossings of $[a, b]$ by $(S_n, n \in \mathbb{N})$.

By the monotone convergence theorem:

$$\mathbb{E}(V_{a,b}) = \lim_{N \rightarrow \infty} \mathbb{E}(V_{a,b,N}) \leq \frac{1}{b-a} (\sup_{N \in \mathbb{N}} \mathbb{E}(S_N^+) + |a|)$$

Since $V_{a,b} \geq 0$, we find that $\mathbb{P}\{V_{a,b} < \infty\} = 1$. Consider the following:

$$\{\liminf_{n \rightarrow \infty} S_n < a < b < \limsup_{n \rightarrow \infty} S_n\} = \{V_{a,b} = \infty\} \text{ therefore } \mathbb{P}\{\liminf_{n \rightarrow \infty} S_n < a < b < \limsup_{n \rightarrow \infty} S_n\} = 0$$

And also:

$$\mathbb{P}\left\{\bigcup_{a < b, a, b \in \mathbb{Q}} \liminf_{n \rightarrow \infty} S_n < a < b < \limsup_{n \rightarrow \infty} S_n\right\} = 0$$

Equivalently, $\mathbb{P}\{\liminf_{n \rightarrow \infty} S_n < \limsup_{n \rightarrow \infty} S_n\} = 0$ and so $\mathbb{P}\{\liminf_{n \rightarrow \infty} S_n = \limsup_{n \rightarrow \infty} S_n\} = 1$.

Therefore (S_n) converges a.s and $S = \lim_{n \rightarrow \infty} S_n$ is the desired r.v.

A priori, $S = \pm\infty$ is possible, but we will exclude this.

Lemma 2.1.9: Fatou's Lemma (probability edition)

Let $X_n \geq 0$ a.s. Then:

$$\mathbb{E}(\liminf_{n \rightarrow \infty} X_n) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n)$$

Here:

$$\mathbb{E}(S^+) = \mathbb{E}(\lim_{n \rightarrow \infty} S_n^+) = \mathbb{E}(\liminf_{n \rightarrow \infty} S_n^+)$$

$$(\text{Fatou}) \implies \leq \liminf_{n \rightarrow \infty} \mathbb{E}(S_n^+) \leq \sup_{n \in \mathbb{N}} \mathbb{E}(S_n^+) < \infty$$

So $\mathbb{E}(S^+) < \infty$. Since $S_n = S_n^+ - S_n^-$ (where $x^- = \max(-x, 0)$), we get $\mathbb{E}(S_n^-) = \mathbb{E}(S_n^+) - \mathbb{E}(S_n) \leq \mathbb{E}(S_n^+) - \mathbb{E}(S_1)$ by the submartingale property, and, by Fatou:

$$\mathbb{E}(S^-) = \mathbb{E}(\liminf_{n \rightarrow \infty} S_n^-) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(S_n^-)$$

$$\leq \liminf_{n \rightarrow \infty} \mathbb{E}(S_n^+) - \mathbb{E}(S_1) \leq \sup_{n \in \mathbb{N}} \mathbb{E}(S_n^+) - \mathbb{E}(S_1) < \infty$$

□

Theorem 2.1.10: L^2 -convergence of L^2 -bounded martingales

Let (S_n) be a martingale relative to (X_n) such that $\sup_{n \in \mathbb{N}} \mathbb{E}(S_n^2) < \infty$. Then there is an r.v S such that

$$\lim_{n \rightarrow \infty} S_n = S \text{ almost surely AND } \lim_{n \rightarrow \infty} \mathbb{E}((S_n - S)^2) = \|S_n - S\|_{L^2}^2 = 0$$

Proof. We have already show that there is an r.v S such that $\lim_{n \rightarrow \infty} S_n = S$ a.s. We must only show that $\lim_{n \rightarrow \infty} \mathbb{E}((S_n - S)^2) = 0$. By the variant on the Doob-Kolmogorov inequality:

$$\mathbb{E}(\max_{i \in \{1, \dots, n\}} S_i^2) \leq 4\mathbb{E}(S_n^2) \text{ and } \max_{i \in \{1, \dots, n\}} \uparrow \sup_{i \in \mathbb{N}} S_i^2 \text{ as } n \rightarrow \infty$$

So, using the monotone convergence theorem:

$$\mathbb{E}(\sup_{i \in \mathbb{N}} S_i^2) = \mathbb{E}(\lim_{n \rightarrow \infty} \max_{i \in \{1, \dots, n\}} S_i^2) = \lim_{n \rightarrow \infty} \mathbb{E}(\max_{i \in \{1, \dots, n\}} S_i^2) \leq 4 \lim_{n \rightarrow \infty} \mathbb{E}(S_n^2) \leq 4 \sup_{n \in \mathbb{N}} \mathbb{E}(S_n^2) < \infty$$

Observe now that: $(S_n - S)^2 = |S_n - S|^2 \leq (|S_n| + |S|)^2 \leq (2 \sup_{i \in \mathbb{N}} |S_i|)^2 = 4 \sup_{i \in \mathbb{N}} S_i^2 = W$, $\mathbb{E}(W) < \infty$, $(S_n - S)^2 \rightarrow 0$ a.s as $n \rightarrow \infty$, and so $(S_n - S)^2 \leq W \forall n$.

By the dominated convergence theorem:

$$\lim_{n \rightarrow \infty} \mathbb{E}((S_n - S)^2) = \mathbb{E}(\lim_{n \rightarrow \infty} (S_n - S)^2) = \mathbb{E}(0) = 0$$

And we get the 'mean squared convergence' we wanted. □

2.2 Branching Processes (Galton-Watson Precesses)

The first model for population growth.

2.2.1 Hypotheses A and B

Definition 2.2.1: Hypothesis A

The numbers of offspring (direct descendants) of the individuals are i.i.d, with the law of a random variable Z , and $\mathbb{P}\{Z = j\} = P_j$, for $j = 0, 1, \dots$

Remark.

$\mathbb{E}(Z)$ is the expected number of offspring per individual. If $\mathbb{E}(Z) > 1$, the population should grow; if $\mathbb{E}(Z) < 1$, the population should dwindle. If $\mathbb{E}(Z) = 1$, what happens...?

First objective: determine the extinction probability = probability that for some generation n , $X_n = 0$ ($\implies X_m = 0 \forall m \geq n$).

Definition 2.2.2: Generating function of Z

The generating function is given (similarly to an MGF) by:

$$g_Z(s) = \mathbb{P}\{Z = 0\} + s\mathbb{P}\{Z = 1\} + s^2\mathbb{P}\{Z = 2\} + \dots = \sum_{j=0}^{\infty} p_j s^j$$

Some properties:

- a) The series converges uniformly on $[-1,1]$.
- b) $s \rightarrow g_Z(s)$ is continuous on $[-1,1]$ and nondecreasing on $[0,1]$.
- c) $g_Z(s) = \mathbb{E}(s^Z)$
- d) If Z and Z' are independent, then $g_{Z+Z'}(s) = g_Z(s) + g_{Z'}(s)$
- e) $g_Z(0) = \mathbb{P}\{Z = 0\}$, $g_Z(1) = 1$; $g'_Z(1) = \mathbb{E}(Z)$ if $\sum_{j=1}^{\infty} jP_j < \infty$, $g''_Z(1) + \mathbb{E}(Z) - (\mathbb{E}(Z))^2 = \text{Var}(Z)$, if $\sum_{j=2}^{\infty} j^2 P_j < \infty$.

Proofs: in problem set 7.

Remark.

The moment generating function of Z is $M_Z(t) = \mathbb{E}(e^{tZ})$ and so we can write $g_Z(s) = M_Z(\ln s)$.

Proposition 2.2.3

Moment generating function of X_n Recall that X_n is the number of individuals in generation n .

$$g_{X_{n+1}}(s) = \mathbb{E}(S^{X_{n+1}}) = \mathbb{E}(\mathbb{E}(S^{X_{n+1}}|X_n)) = \mathbb{E}(\psi(X_n))$$

where we let $\psi(j) = \mathbb{E}(S^{X_{n+1}}|X_n = j)$.

$$\mathbb{E}(S^{X_{n+1}}|X_n = j) = \mathbb{E}(s^{Z_1^n + Z_2^n + \dots + Z_j^n} | X_n = j)$$

And Z_i^n is the number of offspring of an individual i of generation n . Then $\{X_n = j\}$ is determined by Z_p^k , where $k \leq n-1$. By hypothesis A, $\{X_n = j\}$ is independent of the Z_i^n

$$\mathbb{E}(S^{X_{n+1}}|X_n = j) = \mathbb{E}(s^{Z_1^n + \dots + Z_j^n}) = \mathbb{E}(s^{Z_1^n} \dots s^{Z_j^n}) = \mathbb{E}(s^{Z_1^n}) \dots \mathbb{E}(s^{Z_j^n})$$

$$(\text{ident. dist.}) = (\mathbb{E}(s^Z))^j = (g_Z(s))^j \therefore \mathbb{E}(S^{X_{n+1}}|X_n) = \psi(X_n) = (g_Z(s))^{X_n}$$

Then $g_{X_{n+1}}(s) = \mathbb{E}((g_Z(s))^{X_n}) = g_{X_n}(g_Z(s))$, that is

$$g_{X_{n+1}} = g_{X_n} \circ g_Z = (g_{X_{n-1}} \circ g_Z) \circ g_Z \dots \text{ something missing}$$

$$\implies \text{Formula: } g_{X_n} = g_Z^{(n)} = g_Z \circ g_Z \circ \dots \text{ (n times) } \circ g_Z$$

$$\implies g_Z(s) = \sum_{j=0}^{\infty} s^j \mathbb{P}\{Z = j\} = \mathbb{E}(s^Z)$$

Definition 2.2.4: Extinction probability.

Let $F = \bigcup_{n=1}^{\infty} \{X_n = 0\}$ = "the population becomes extinct". Find $\mathbb{P}(F)$... how? Observe that $\{X_n = 0\} \subset \{X_{n+1} = 0\}$ so $\mathbb{P}(F) = \lim_{n \rightarrow \infty} \mathbb{P}\{X_n = 0\}$.

Theorem 2.2.5

Suppose that $X_0 = 1$ and that Hypothesis A holds. Then $\mathbb{P}(F)$ is the smallest number $\alpha \geq 0$ such that $\alpha = g_Z(\alpha) = \sum_{j=0}^{\infty} p_j \alpha^j$.

Remark.

If $p_0 = 0 = \mathbb{P}\{Z = 0\}$ (= prob. that an individual has no offspring) then $\mathbb{P}(F) = 0$ and $0 = g_Z(0) = p_0 + p_1(0) + p_2(0^2) + \dots = 0$

Proof. Of the theorem. Set $\alpha_n = \mathbb{P}\{X_n = 0\}$. Write $\mathbb{P}(F) = \lim_{n \rightarrow \infty} \mathbb{P}\{X_n = 0\} = \lim_{n \rightarrow \infty} \alpha_n$. Now:

$$\begin{aligned} \alpha_n &= \mathbb{P}\{X_n = 0\} = g_{X_n}(0) = g_Z^{(n)}(0) = (g_Z \circ g_{X_{n-1}})(0) \\ &= g_Z(g_{X_{n-1}}(0)) = g_Z(\alpha_{n-1}) \end{aligned}$$

Let $n \rightarrow \infty$. $\alpha = \lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} g_Z(\alpha_{n-1})$; since it's continuous, $= g_Z(\lim_{n \rightarrow \infty} \alpha_{n-1}) = g_Z(\alpha)$.

To prove that $\alpha = \mathbb{P}(F)$ is the **smallest nonnegative solution**:

Let β be another solution: $\beta \geq 0$ and $\beta = g_Z(\beta)$. g_Z is nondecreasing on $[0, 1]$, so since $\beta \geq 0$, $g_Z(0) \leq g_Z(\beta) = \beta$

$$\implies g_Z(g_Z(0)) \leq g_Z(\beta) = \beta$$

$$g_Z^{(2)} \leq \beta$$

$\vdots$

$$\alpha_n = g_Z^{(n)}(0) \leq \beta \text{ so as } n \rightarrow \infty, \alpha \leq \beta.$$

□

Definition 2.2.6: Hypothesis B

a) $p_j < 1 \forall j$, and $p_0 + p_1 < 1$ (if $p_0 + p_1 = 1$, then $p_j = 0$ for $j \geq 2$, each has 0 or 1 offspring, nothing interesting happens $\implies \text{Var}(Z) > 0$.)

b) $\mathbb{E}(Z) < \infty$

Proposition 2.2.7

Under hypotheses A and B:

a) prob. extinction = 1 $\iff \mathbb{E}(Z) \leq 1$.

b) prob. extinction = 0 $\iff p_0 = \mathbb{P}\{Z = 0\} = 0$.

Proof. Observations: $g_Z(1) = 1, g_Z(0) = p_0, g'_Z(1) = \sum_{j=1}^{\infty} j p_j = \mathbb{E}(Z), g''_Z(s) = \sum_{j=2}^{\infty} j(j-1)s^{j-2} p_j > 0$ for $s > 0$.

So $s \rightarrow g_Z(s)$ is strictly convex on $[0, 1]$ and nondecreasing.

□

2.2.2 Asymptotic behaviour of X_n as $n \rightarrow \infty$

Lemma 2.2.8

Set $m = \mathbb{E}(Z)$, $\sigma^2 = \text{Var}(Z) < \infty$. Then:

$$\mathbb{E}(X_n) = m^n, \quad \text{Var}(X_n) = \begin{cases} \sigma^2 \frac{m^n(m^n-1)}{m^2-m} & m \neq 1 \\ n\sigma^2 & m = 1 \end{cases}$$

Proof.

$$\mathbb{E}(X_n) = g'_{X_n}(1) = (g_{X_{n-1}} \circ g_Z)'(1) = g'_{X_{n-1}}(g_Z(1)) \cdot g'_Z(1) = mg'_{X_{n-1}}(1)$$

$$\mathbb{E}(X_n) = m\mathbb{E}(X_{n-1}) = m^2\mathbb{E}(X_{n-2}) = \dots = m^n\mathbb{E}(X_0) = m^n$$

Recall: $g''_{X_n} = \text{Var}(X_n) - \mathbb{E}(X_n) + (\mathbb{E}(X_n))^2$.

(This proof is unfinished.) □

Theorem 2.2.9

Let $S_n = \frac{X_n}{m^n}$. Then (S_n) is a martingale relative to (X_n) . If $m > 1$, there exists an r.v S such that $S_n \rightarrow S$ a.s and $\mathbb{E}((S_n - S)^2) \rightarrow 0$ as $n \rightarrow \infty$.

Proof.

$$\mathbb{E}(|S_n|) = \mathbb{E}(S_n) = \mathbb{E}\left(\frac{X_n}{m^n}\right) = \frac{\mathbb{E}(X_n)}{m^n} = 1 < \infty$$

Next:

$$\mathbb{E}(S_{n+1}|X_1, \dots, X_n) = \mathbb{E}\left(\frac{X_{n+1}}{m^{n+1}}|X_1, \dots, X_n\right) = \frac{1}{m^{n+1}}\mathbb{E}(X_{n+1}|X_1, \dots, X_n)$$

Observe that $\mathbb{E}(X_{n+1}|X_n = j, X_{n-1} = j_{n-1}, \dots, X_1 = j_1) = (\text{Hyp. A}) = \mathbb{E}(Z_1^n + \dots + Z_j^n) = j\mathbb{E}(Z) = mj$

So $\mathbb{E}(X_{n+1}|X_1, \dots, X_n) = mX_n$, and $\mathbb{E}(S_{n+1}|X_1, \dots, X_n) = \frac{1}{m^{n+1}}mX_n = \frac{X_n}{m^n} = S_n$.

Now, to use the Martingale convergence theorem for L^2 -bounded martingales, we suppose $m > 1$:

$$\begin{aligned} \mathbb{E}(S_n^2) &= \text{Var}(S_n) + (\mathbb{E}(S_n))^2 = \frac{\sigma^2}{m^{2n}} \frac{m^n(m^n-1)}{m^2-m} + 1 \\ &= \frac{\sigma^2}{m^2-m} \left(1 - \frac{1}{m^n}\right) + 1 \leq \frac{\sigma^2}{m^2-m} + 1 < \infty \end{aligned}$$

Therefore, $\exists$ an r.v S such that $S_n \rightarrow S$ a.s, and $\mathbb{E}((S_n - S)^2) \rightarrow 0$ as $n \rightarrow \infty$. □

There are some conclusions we can draw:

Remark.

In the end:

- a) If $m \leq 1$, then $\mathbb{P}\{\exists n : X_n = 0\} = 1$.
 - b) If $m > 1$, then for large n , $X_n \sim S \cdot m^n$. $\mathbb{P}\{S \geq 0\} = 1$. If $S > 0$, the population grows exponentially. If $S = 0$, $\frac{X_n}{m^n} \rightarrow 0$; the population "grows" more slowly than m^n ; in fact we will show that the population become extinct in this case.
- So it is either explosion or extinction (under Hypothesis B).

Proposition 2.2.10

If $m > 1$:

- 1) $\mathbb{P}\{S = 0\} = \mathbb{P}\{\text{extinction}\}$.
- 2) $\text{Var}(S|S > 0) > 0$: given S positive (no extinction), S is still random.

Proof. Proof of 1): Clearly $\{\text{extinction}\} \subset \{S = 0\}$ because $S_n = \frac{X_n}{m^n} \rightarrow S$ a.s, so $\mathbb{P}\{\text{extinction}\} \leq \mathbb{P}\{S = 0\}$.

To check that inequality is an equality: let $q = \mathbb{P}\{S = 0\}$. Notice that $\{S = 0\} = \{\lim_{n \rightarrow \infty} S_n = 0\}$.

Then, if $X_n^{(\ell)} = \#$ of individuals in generation n that descend from individual ℓ in generation 1:

$$\begin{aligned} \mathbb{P}\left\{\lim_{n \rightarrow \infty} S_n = 0 | X_1 = k\right\} &= \mathbb{P}\left\{\lim_{n \rightarrow \infty} \frac{1}{m^n} \sum_{\ell=1}^k X_n^{(\ell)} | X_1 = k\right\} \\ &= \mathbb{P}\left(\bigcap_{\ell=1}^{\infty} \left\{\lim_{n \rightarrow \infty} \frac{X_n^{(\ell)}}{m^n} = 0\right\} \mid X_1 = k\right) = (\text{Hyp. A}) = \prod_{\ell=1}^k \mathbb{P}\left\{\lim_{n \rightarrow \infty} \frac{X_n^{(\ell)}}{m^n} = 0 \mid X_1 = k\right\} \\ &= \prod_{\ell=1}^k \mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{X_n}{m^n} = 0\right) = (\mathbb{P}\{S = 0\})^k \end{aligned}$$

Therefore:

$$q = \mathbb{P}\{S = 0\} = \sum_{k=0}^{\infty} \mathbb{P}\{S = 0 \mid X_n = k\} \mathbb{P}\{X_1 = K\} = \sum_{k=0}^{\infty} q^k p_k = g_Z(q)$$

That is: $q = g_Z(q)$. In addition, $q = \mathbb{P}\{S = 0\} < 1$ because $\text{Var}(S) > 0$. We know that $s = g_Z(s)$ has exactly two solutions ≥ 0 when $m > 1$, namely 1 and $\mathbb{P}\{\text{extinction}\}$, and so $q = \mathbb{P}\{\text{extinction}\}$. $\square$

Chapter 3: Brownian motion

”As the normal distribution is to distributions of data, Brownian motion is to evolution in continuous time.”

Definition 3.0.1: Brownian motion

A Brownian motion is a stochastic process $(B_t, t \in \mathbb{R}_+)$ with two properties:

$$\text{a) } \forall s, t \in \mathbb{R}_+, B_{t+s} - B_s \sim N(0, \sigma^2 t) \quad (\sigma > 0 \text{ fixed})$$

b) For $t_1 < t_2 \leq t_3 < t_4$, the increments $B_{t_4} - B_{t_3}$ and $B_{t_2} - B_{t_1}$ are independent (and the same holds for increments over n non-overlapping intervals).

Remark.

The law of B_0 is not specified. Unless otherwise specified, we take $\sigma = 1$ and $B_0 \equiv 0$: we call this *standard* BM (since $B_t \sim N(0, t)$).

Proposition 3.0.2

Joint probability distribution of $(B_{t_n}), t_1 < t_2 < \dots < t_n$. Define:

$$p(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}, \quad \text{the density of } N(0, 1)$$

Then $f_{(B_{t_1}, \dots, B_{t_n})}(x_1, \dots, x_n) = p(x_1, t_1)p(x_2 - x_1, t_2 - t_1) \dots p(x_n - x_{n-1}, t_n - t_{n-1})$

Proof. The probability density function of $Y = (B_{t_1} - B_0, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}})$ is $g(y_1, \dots, y_n) = p(y_1, t_1 - 0)p(y_2, t_2 - t_1) \dots p(y_n, t_n - t_{n-1})$.

Define $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $T(x_1, \dots, x_n) = (x_1, x_2 - x_1, \dots, x_n - x_{n-1})$.

Then $T(B_{t_1}, \dots, B_{t_n}) = Y$.

Recall from probability that if $Y = T(X)$, T bijective, C^0 , then $f_X(x) = f_Y(T(x))|\det J|$ where J is the Jacobian. In this case, the entries J_{ii} of the Jacobian are all 1, with the

entries $J_{i+1,i} = -1$. In any case, $\det J = 1$.

Therefore, $f_{(B_{t_1}, \dots, B_{t_n})}(x_1, \dots, x_n) = p(x_1, t_1)p(x_2 - x_1, t_2 - t_1) \dots p(x_n - x_{n-1}, t_n - t_{n-1})$. $\square$

Proposition 3.0.3

The joint density of (B_s, B_t) is:

$$f(y, x) = p(y, s)p(x - y, t - s)$$

Conditional density of B_t given $B_s = y$, for $s < t$:

$$f_{B_t}(x \mid B_s = y) = \frac{f(y, x)}{f_{B_s}} = \frac{p(y, s)p(x - y, t - s)}{p(y, s)} = p(x - y, t - s)$$

It follows that:

$$\mathbb{P}\{B_t \leq x \mid B_s = y\} = \int_{-\infty}^x p(x - y, t - s) dv \equiv \int_{-\infty}^{x - y} p(u, t - s) du = \mathbb{P}\{B_t \leq x - y \mid B_s = 0\}$$

Which implies that Brownian motion is **invariant under translations of space**.

Proposition 3.0.4

Continuity of sample paths, to be proved later:

With probability 1:

$$t \rightarrow B_t(\omega), \omega \in \Omega \text{ fixed}; \mathbb{R}^+ \rightarrow \mathbb{R}$$

is continuous.

Proposition 3.0.5

Covariance of B_s and B_t :

$$\mathbb{E}(B_s B_t) = \min(s, t) = s \wedge t$$

Proof. Let $s < t$. Then $\mathbb{E}(B_s B_t) = \mathbb{E}(B_s(B_s + B_t - B_s)) = \mathbb{E}(B_s^2) + \mathbb{E}(B_s(B_t - B_s)) = s + \mathbb{E}(B_s)\mathbb{E}(B_t - B_s) = s + 0 \cdot 0 = s = \min(s, t)$. $\square$

3.1 Behaviours of Brownian motion

(A line is missing here.)

Remark.

Many interesting events are **not** of this kind. Example:

$$\mathbb{P} \left\{ \max_{u \in [0, t]} B_u > a \right\} = \mathbb{P} \left(\bigcup_{r \in \mathbb{Q} \cap [0, t]} \{B_r > a\} \right)$$

which is not a disjoint union.

Proposition 3.1.1

Reflection principle; for $a \geq 0$:

$$\mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a \right\} = 2\mathbb{P}\{B_t \geq a\} = 2 \int_a^\infty \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx$$

Proof. (Explanation). Let $\tau = \inf\{u \geq 0 : B_u = a\}$ if $\{\dots\} \neq \emptyset$ and ∞ otherwise. Define:

$$\tilde{B}_t = \begin{cases} B_t & \text{if } t < \tau \\ a - (B_t - a) & \text{if } t \geq \tau \end{cases}$$

By "symmetry". ($\tilde{B}_t$ is also a Brownian motion (though for a rigorous justification we need the strong Markov property). Consider the event:

$$\left\{ \max_{u \in [0, t]} B_u \geq a, B_t \geq a \right\} = \left\{ \max_{u \in [0, t]} \tilde{B}_u \geq a, \tilde{B}_t \leq a \right\}$$

Therefore $\mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a, B_t \geq a \right\} = \mathbb{P} \left\{ \max_{u \in [0, t]} \tilde{B}_u \geq a, \tilde{B}_t \leq a \right\}$ which, since both B and $\tilde{B}$ are Brownian motions, $= \mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a, B_t \leq a \right\}$. So:

$$\mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a \right\} = \mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a, B_t \geq a \right\} + \mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a, B_t \leq a \right\}$$

(note that $\mathbb{P}\{B_t = a\} = 0$)

$$= 2\mathbb{P} \left\{ \max_{u \in [0, t]} B_u \geq a, B_t \geq a \right\} = 2\mathbb{P}\{B_t \geq a\}$$

□

Definition 3.1.2: First hitting time of a level

Fix $a \neq 0$, and set $T_a = \inf\{t \geq 0 : B_t = a\}$ if $\{\dots\} \neq \emptyset$ and ∞ otherwise. Then the probability density function of T_a is:

$$f_{T_a}(T) = \frac{|a|}{\sqrt{2\pi}} \frac{1}{\sqrt{t^3}} e^{-\frac{a^2}{2t}}, \quad \text{for } t > 0$$

Proof. (Justification) Suppose $a > 0$. Then $\{T_a \leq t\} = \{\max_{u \in [0,t]} B_u \geq a\}$, so, by the reflection principle:

$$\begin{aligned} \mathbb{P}\{T_a \leq t\} &= \mathbb{P}\left\{\max_{u \in [0,t]} B_u \geq a\right\} = 2\mathbb{P}\{B_t \geq a\} = 2 \int_a^\infty \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx \\ &= \frac{2}{\sqrt{2\pi t}} \int_{\frac{a}{\sqrt{t}}}^\infty e^{-\frac{y^2}{2}} \sqrt{t} dy = \frac{2}{\sqrt{2\pi}} \int_{\frac{a}{\sqrt{t}}}^\infty e^{-\frac{y^2}{2}} dy \end{aligned}$$

Then:

$$f_{T_a}(t) = \frac{d}{dt} \mathbb{P}\{T_a \leq t\} = \frac{2}{\sqrt{2\pi}} \left(-e^{-\frac{a^2}{2t}}\right) \left(-\frac{1}{2} a t^{-\frac{3}{2}}\right) = \frac{a}{\sqrt{2\pi}} t^{-\frac{3}{2}} e^{-\frac{a^2}{2t}}$$

□

Remark.

- a) $\mathbb{P}\{T_a < \infty\} = \lim_{t \uparrow \infty} \mathbb{P}\{T_a \leq t\} = \lim_{t \rightarrow \infty} \frac{2}{\sqrt{2\pi}} \int_{\frac{a}{\sqrt{t}}}^\infty e^{-\frac{y^2}{2}} dy = \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-\frac{y^2}{2}} dy = 1$
- b) $\mathbb{E}(T_a) = \int_0^\infty t \cdot f_{T_a}(t) dt = \int_0^\infty t \frac{a}{\sqrt{2\pi}} t^{-\frac{3}{2}} \exp\left(-\frac{a^2}{2t}\right) dt \geq \int_1^\infty \frac{a}{\sqrt{2\pi}} t^{-\frac{1}{2}} \exp\left(-\frac{a^2}{2t}\right) dt = \infty$
(similar to a simple random walk).

Definition 3.1.3: Probability of moving down a units

What is $\mathbb{P}\{\min_{u \in [0,t]} B_u \leq 0 \mid B_0 = a\}$?

By symmetry:

$$\mathbb{P}\left\{\min_{u \in [0,t]} B_u \leq 0 \mid B_0 = 0\right\} = \mathbb{P}\left\{\max_{u \in [0,t]} B_u \geq 2a \mid B_0 = a\right\} = \mathbb{P}\left\{\max_{u \in [0,t]} B_u \geq a \mid B_0 = 0\right\}$$

By invariance under translation. Then:

$$= \mathbb{P}\{T_a \leq t \mid B_0 = 0\} = \frac{a}{\sqrt{2\pi}} \int_0^t u^{-\frac{3}{2}} \exp\left(-\frac{a^2}{2u}\right) du$$

Proposition 3.1.4

Probability of visiting zero during $[t_0, t_1]$

$$\begin{aligned} \mathbb{P} \left\{ \min_{u \in [t_0, t_1]} B_u \leq 0 \mid B_{t_0} = a \right\} &= \mathbb{P} \left\{ \min_{u \in [0, t_1 - t_0]} B_u \leq 0 \mid B_0 = a \right\} \\ &= \frac{a}{\sqrt{2\pi}} \int_0^{t_1 - t_0} u^{-\frac{3}{2}} \exp \left(-\frac{a^2}{2u} \right) du \end{aligned}$$

Proposition 3.1.5

Probability of visiting zero during (t_0, t_1)

We compute $\alpha = \mathbb{P}\{\exists t \in (t_0, t_1) : B_t = 0 \mid B_0 = 0\} = \frac{2}{\pi} \arccos \left(\sqrt{\frac{t_0}{t_1}} \right)$
by an exercise mostly in integration.

Proposition 3.1.6

The arcsin law (last visit to 0 before time 1)

Let $L_1 = \sup\{t \leq 1 : B_t = 0\}$ = time of last visit to 0 before time 1.

Note: $\mathbb{P}\{B_1 = 0\} = 0, \mathbb{P}\{L_1 < 1\} = 1$. Then for $s \in (0, 1)$:

$$\mathbb{P}\{L_1 \leq s\} = \frac{2}{\pi} \arcsin(\sqrt{s})$$

Proof. $\{L_1 \leq s\} = \{B_t \neq 0 \text{ for all } t \in [s, 1]\}$, so

$$\mathbb{P}\{L_1 \leq s\} = 1 - \mathbb{P}\{\exists t \in [s, 1] : B_t = 0 \mid B_0 = 0\} = 1 - \frac{2}{\pi} \arccos(\sqrt{s})$$

and the result follows from Prop. 3.1.5. □

Remark.

The probability density function of L_1 :

$$f_{L_1}(s) = \frac{2}{\pi} \frac{1}{\sqrt{1-s}} \frac{1}{2\sqrt{s}} = \frac{1}{\pi \sqrt{s(1-s)}}, \quad s \in (0, 1)$$

3.2 Transformations of Brownian motion

Remark.

Notation: $P_a\{\dots\} = \mathbb{P}\{\dots \mid B_0 = a\}$

Those that preserve the properties:

Proposition 3.2.1

Scaling invariance:

Let $(B_t, t \in \mathbb{R}_+)$ be a standard Brownian motion. Fix $a > 0$ and define $\tilde{B}_t = \frac{1}{\sqrt{a}}B_{at}$. Then $(\tilde{B}_t)$ is a standard Brownian motion.

Proof. We check the two defining properties of Brownian motion:

$$\text{a) } \tilde{B}_{t+s} - \tilde{B}_s = \frac{1}{\sqrt{a}}B_{a(t+s)} - \frac{1}{\sqrt{a}}B_{as} = \frac{1}{\sqrt{a}}(B_{as+at} - B_{as})$$

$$(\text{BW}) = \frac{1}{\sqrt{a}}N(0, at) = N(0, t) \text{ [law]}$$

$$\text{b) } \tilde{B}_{t_4} - \tilde{B}_{t_3} = \frac{1}{\sqrt{a}}(B_{at_4} - B_{at_3})$$

$$\tilde{B}_{t_2} - \tilde{B}_{t_1} = \frac{1}{\sqrt{a}}(B_{at_2} - B_{at_1})$$

Then these are independent because $t_1 < t_2 \leq t_3 < t_4 \implies at_1 < at_2 \leq at_3 < at_4$. □

Proposition 3.2.2

Time inversion:

Let (B_t) be a standard Brownian motion. Define $\tilde{B}_t = tB_{1/t}$ if $t > 0$, and $\tilde{B}_0 = 0$. Then $(\tilde{B}_t)$ is a standard Brownian motion.

Proof. We check the 2 properties:

$$\text{a) } \tilde{B}_{t+s} - \tilde{B}_s = (t+s)B_{\frac{1}{t+s}} - sB_{1/s} = tB_{\frac{1}{t+s}} - s(B_{1/s} - B_{\frac{1}{t+s}})$$

$$(\text{law}) = tN(0, \frac{1}{t+s}) - sN(0, \frac{1}{s} - \frac{1}{t+s})$$

$$= N(0, t^2 \frac{1}{t+s} + s^2(\frac{1}{s} - \frac{1}{t+s})) = N(0, t)$$

$$\text{b) } \tilde{B}_{t_4} - \tilde{B}_{t_3} = t_4 B_{1/t_4} - t_3 B_{1/t_3}; \quad \tilde{B}_{t_2} - \tilde{B}_{t_1} = t_2 B_{1/t_2} - t_1 B_{1/t_1}$$

For $t_1 < t_2 \leq t_3 < t_4$.

To check independence, we compute the covariance (using $\mathbb{E}(B_u B_v) = u \wedge v$):

$$\begin{aligned} \mathbb{E}((\tilde{B}_{t_4} - \tilde{B}_{t_3})(\tilde{B}_{t_2} - \tilde{B}_{t_1})) &= t_4 t_2 \mathbb{E}(B_{1/t_4} B_{1/t_2}) - t_4 t_1 \mathbb{E}(B_{1/t_4} B_{1/t_1}) - t_3 t_2 \mathbb{E}(B_{1/t_3} B_{1/t_2}) + t_3 t_1 \mathbb{E}(B_{1/t_3} B_{1/t_1}) \\ &= t_4 t_2 \frac{1}{t_4} - t_4 t_1 \frac{1}{t_4} - t_3 t_2 \frac{1}{t_3} + t_3 t_1 \frac{1}{t_3} = 0. \end{aligned}$$

□

Remark.

The continuity of $t \rightarrow \tilde{B}_t(w)$ at $t = 0$ is obvious. It is suggested by the fact that:

$$\mathbb{E}((\tilde{B}_t)^2) = t^2 \mathbb{E}(B_{1/t}^2) = t^2 \frac{1}{t} = t \rightarrow 0$$

Proposition 3.2.3

Invariance under translations of time:

Fix $h > 0$. Define $\tilde{B}_t = B_{t+h} - B_h$. Then $(\tilde{B}_t)$ is a standard Brownian motion.

Proof. We check the 2 properties:

$$\begin{aligned} \text{a) } \tilde{B}_{t+s} - \tilde{B}_s &= (B_{t+s+h} - B_h) - (B_{s+h} - B_h) \\ &= B_{t+s+h} - B_{s+h} = (\text{law}) N(0, t) \end{aligned}$$

$$\text{b) } \tilde{B}_{t_4} - \tilde{B}_{t_3} = B_{t_4+h} - B_{t_3+h}; \quad \tilde{B}_{t_2} - \tilde{B}_{t_1} = B_{t_2+h} - B_{t_1+h}$$

Which are independent since $t_1 < t_2 \leq t_3 < t_4 \implies t_1 + h < t_2 + h \leq t_3 + h < t_4 + h$ □

Those that **do not** preserve the properties:

Definition 3.2.4: Reflected Brownian motion

A reflected Brownian motion is a process $(X_t, t \in \mathbb{R}_+)$ defined by $X_t = |B_t|$, where (B_t) is a standard Brownian motion.

”The particle of pollen bounced off of the floor of the basin.”

Proposition 3.2.5

Reflected Brownian motion has the Markov property.

In addition, for $0 \leq s < t$:

$$f_{X_t}(y \mid X_s = x) = p(y - x, t - s) + p(y + x, t - s) \quad y, x \geq 0$$

(where $p(u, r) = \frac{1}{\sqrt{2\pi r}} \exp(-\frac{u^2}{2r})$)

Proof. Fix $0 \leq t_1 < t_2 < \dots < t_n < t$ and $x_1, \dots, x_n, x \geq 0$. Then:

$$\begin{aligned} & \mathbb{P}\{X_t \leq x \mid X_{t_n} = x_n, X_{t_{n-1}} = x_{n-1}, \dots, X_{t_1} = x_1\} \\ &= \mathbb{P}\{-x \leq B_t \leq x \mid B_{t_n} = \pm x_n, B_{t_{n-1}} = \pm x_{n-1}, \dots, B_{t_1} = \pm x_1\} \\ &= \mathbb{P}\{-x \leq B_t \leq x \mid (B_{t_n} = x_n, \text{past}) \text{ or } (B_{t_n} = -x_n, \text{past})\} \\ &= \mathbb{P}\{-x \leq B_t \leq x \mid B_{t_n} = x_n, \text{past}\} \end{aligned}$$

but since the original Brownian motion has the Markov property:

$$= \mathbb{P}\{-x \leq B_t \leq x \mid B_{t_n} = x_n\} = \int_{-x}^x p(y - x_n, t - t_n) dy$$

$= \mathbb{P}\{X_t \leq x \mid X_{t_n} = x_n\}$; the past positions did not affect the calculation. Then the conditional density $\frac{d}{dx} \mathbb{P}\{X_t \leq x \mid X_{t_n} = x_n\}$ comes out as described above. $\square$

Definition 3.2.6: Absorbed Brownian motion

Let (B_t) be a Brownian motion with $B_0 = a \neq 0$. Then the Brownian motion absorbed at 0 is the process $(Y_t, t \in \mathbb{R}_+)$ defined by $Y_t = B_t$ if $t \leq T_0$ where $T_0 = \inf\{t \geq 0 : B_t = 0\}$, and $Y_t = 0$

"The particle of pollen sticks to the floor of the basin."

Proposition 3.2.7

Brownian motion absorbed at 0 has the Markov property. In addition, for $0 \leq t_n < t$:

$$f_{Y_t}(y \mid Y_{t_n} = x) = p(y - x, t - t_n) - p(y + x, t - t_n) \quad x, y > 0$$

Proof. Let $a > 0$. We want to see if:

$$\mathbb{P}\{Y_t > y \mid Y_{t_n} = x, Y_{t_{n-1}} = x_{n-1}, \dots, Y_{t_1} = x_1\} = \mathbb{P}\{Y_t > y \mid Y_{t_n} = x\}$$

With $0 \leq t_1 < t_2 < \dots < t_n < t$ and $y \geq 0, x \geq 0, x_i > 0, i = 1, \dots, n-1$.

If $x = 0$, then both sides are equal to 0. If $x > 0$, then the LHS is:

$$\mathbb{P}\{Y_t > y \mid \min_{0 \leq u \leq t_n} B_u > 0, Y_{t_n} = x, Y_{t_{n-1}} = x_{n-1}, \dots, Y_{t_1} = x_1\}$$

(rest of proof missing, allergies)

□

Proposition 3.2.8

Probability of being absorbed at 0 before time t :

$$\mathbb{P}\{Y_t = 0 \mid Y_0 = x\} = 1 - \mathbb{P}\{Y_t > 0 \mid Y_0 = x\} = 1 - \int_{-x}^x p(u, t - t_n) du > 0$$

(Y_t is neither a discrete nor a continuous r.v.)

Definition 3.2.9: Brownian motion with drift

This is the process $(Z_t, t \in \mathbb{R}_+)$ defined by $Z_t = B_t + \mu t$ where (B_t) is a standard Brownian motion, and $\mu \neq 0$.

”You can imagine that there’s a current moving along at speed μ ...”

Remark.

Independence of increments for Brownian motion with drift

For $0 \leq t_1 < t_2 \leq t_3 < t_4$:

$$Z_{t_4} - Z_{t_3} = B_{t_4} - B_{t_3} + \mu(t_4 - t_3); \quad Z_{t_2} - Z_{t_1} = B_{t_2} - B_{t_1} + \mu(t_2 - t_1)$$

Then:

$$f_{Z_{t_2}-Z_{t_1}, Z_{t_4}-Z_{t_3}}(x, y) = f_{B_{t_2}-B_{t_1}}(x - \mu(t_2 - t_1)) f_{B_{t_4}-B_{t_3}}(y - \mu(t_4 - t_3))$$

Proposition 3.2.10

Brownian motion with drift has the Markov property

Proof. Fix $0 \leq t_1 < \dots < t_n < t$. Then

$$P_0\{Z_t \leq x \mid Z_{t_n} = x_n, \dots, Z_{t_1} = x_1\}$$

$$= \mathbb{P}\{Z_t - Z_{t_n} \leq x - x_n \mid Z_{t_n} - Z_{t_{n-1}} = x_n - x_{n-1}, \dots, Z_{t_2} - Z_{t_1} = x_2 - x_1, Z_{t_1} - Z_0 = x_1 - 0\}$$

which by independence

$$= \mathbb{P}\{Z_t - Z_{t_n} \leq x - x_n\} = \mathbb{P}\{Z_t - Z_{t_n} \leq x - x_n \mid Z_{t_n} - Z_0 = x_n - 0\} = \mathbb{P}\{Z_t \leq x \mid Z_{t_n} = x_n\}$$

□

3.3 Exit times

Proposition 3.3.1

Time to exit an interval for standard Brownian motion

Define, for $a \in \mathbb{R}$: $\tau_a = \inf\{u \geq 0 : B_t = a\}$. Then $\mathbb{P}\{T_a < \infty\} = 1$.

Let $a < b$, then define $T_{a,b} = \tau_a \wedge \tau_b$. If $B_0 = x \in [a, b]$, then $T_{a,b}$ is by definition the first exit time of $[a, b]$.

So for $a < x < b$: fix $h > 0$ (small) such that $[x - h, x + h] \subset (a, b)$. Then Brownian motion starting at x must exit $[x - h, x + h]$ first before being able to exit $[a, b]$. By symmetry, $P_x\{B_{T_{x-h, x+h}} = x + h\} = \frac{1}{2} = P_x\{B_{T_{x-h, x+h}} = x - h\}$. Therefore, by the law of total probability:

$$\begin{aligned} P_x\{B_{T_{a,b}} = b\} &= P_x\{B_{T_{a,b}} = b \mid B_{T_{x-h, x+h}} = x + h\} \cdot P_x\{B_{T_{x-h, x+h}} = x + h\} \\ &\quad + P_x\{B_{T_{a,b}} = b \mid B_{T_{x-h, x+h}} = x - h\} \cdot P_x\{B_{T_{x-h, x+h}} = x - h\} \end{aligned}$$

Let $f(x) = P_x\{B_{T_{a,b}} = b\}$; then $f(x) = \frac{1}{2}f(x + h) + \frac{1}{2}f(x - h)$.

This implies that f is affine, so if f is C^2 :

$$\frac{1}{2}[f(x + h) - 2f(x) + f(x - h)] = 0 = \frac{\frac{1}{2}[f(x + h) - 2f(x) + f(x - h)]}{h^2}$$

So as $h \downarrow 0$, we find that $\frac{1}{2}f''(x) = 0 \implies f(x) = \alpha x + \beta$; $f(a) = 0, f(b) = 1$.

Proposition 3.3.2

For $a \leq x \leq b$:

$$P_x\{B_{T_{a,b}} = b\} = \frac{x - a}{b - a}$$

Proposition 3.3.3

Mean exit time of $[a, b]$. Problem: determine $E_x(T_{a,b}), a \leq x \leq b$.

(Recall: $E_x(\tau_a) = \infty = E_x(\tau_b), T_{a,b} = \tau_a \wedge \tau_b$).

Fix $h > 0$. We seek $E_x(T_{x-h, x+h}) = E_0(T_{-h, h})$.

$T_{-h, h} = \inf\{u \geq 0 : |B_u| = h\}$.

By the scaling invariance property, $(\tilde{B}_u = hB_{\frac{u}{h^2}}, u \in \mathbb{R}_+)$ is also a standard Brownian motion. Therefore, $T_{-h, h}$ has the same law as $\tilde{T}_{-h, h} = \inf\{u \geq 0 : |\tilde{B}_u| = h\}$.

$$\begin{aligned} \tilde{T}_{-h, h} &= \inf\{u \geq 0 : |\tilde{B}_u| = h\} = \inf\{u \geq 0 : \left| hB_{\frac{u}{h^2}} \right| = h\} \\ &= \inf\{u \geq 0 : \left| B_{\frac{u}{h^2}} \right| = 1\} = \inf\{vh^2 \geq 0, |B_v| = 1\} \\ &= h^2 \inf\{v \geq 0 : |B_v| = 1\} = h^2 T_{-1, 1} \end{aligned}$$

So $E_0(T_{-h, h})$, by equality in law, $= E_0(\tilde{T}_{-h, h}) = h^2 E_0(T_{-1, 1}) = c_0 h^2$

Calculation of $E_X(T_{a,b})$: Fix $h > 0$, small. The Brownian motion starting from x must first exit $[x-h, x+h]$. For this, it takes $c_0 h^2$ units of time, then, starting from $x+h$ or $x-h$, it must exit $[a, b]$. So:

$$E_x(T_{a,b}) = (c_0 h^2 + E_{x-h}(T_{a,b})) \frac{1}{2} + (c_0 h^2 + E_{x+h}(T_{a,b})) \frac{1}{2}$$

Let $g(x) = E_x(T_{a,b})$. Then:

$$\begin{aligned} g(x) &= (c_0 h^2 + g(x-h)) \frac{1}{2} + (c_0 h^2 + g(x+h)) \frac{1}{2} \\ \implies c_0 h^2 + \frac{1}{2}(g(x-h) - 2g(x) + g(x+h)) &= 0 \end{aligned}$$

Divide by h^2 , then as long as g is C^2 , when $h \downarrow 0$:

$$0 = c_0 + \frac{1}{2} \frac{g(x-h) - 2g(x) + g(x+h)}{h^2} \implies 0 = c_0 + \frac{1}{2} g''(x)$$

So $g''(x) = -2c_0$ and g is a 2nd degree polynomial; $g(a) = 0 = g(b), g(x) = c_0(b-x)(x-a)$.

Proposition 3.3.4

For $a \leq x \leq b$:

$$E_x(T_{a,b}) = (b - a)(x - a)$$

In particular, $c_0 = 1$ (but this is very hard to prove so we will not).

Proposition 3.3.5

Exit time for Brownian motion with drift

We write (as before) $Z_t = B_t + \mu t$, $\mu \neq 0$, for B_t a standard Brownian motion. Define $T_{a,b} = \inf\{u \geq 0 : Z_u = a \text{ or } Z_u = b\}$.

Problem: we want to evaluate $P_x\{Z_{T_{a,b}} = b\}$.

Fix $h > 0$, small. During h^2 units of time (variance), the particle moves about $\pm h$ (standard dev.) units. The drift contributes $\mu h^2 \ll h$ since h is small.

Must have:

$$P_x\{Z_{T_{a,b}} = b\} = \frac{1}{2}P_{x+\mu h^2+h}\{Z_{T_{a,b}} = b\} + \frac{1}{2}P_{x+\mu h^2-h}\{Z_{T_{a,b}} = b\}$$

Let $f(x) = P_x\{Z_{T_{a,b}} = b\}$. Then $f(x) = \frac{1}{2}f(x + \mu h^2 + h) + \frac{1}{2}f(x + \mu h^2 - h)$

[Some really insane steps here; but the usual, something then divide by h^2 , and as $h \downarrow 0$:]

$0 = \mu f'(x) + \frac{1}{2}f''(x)$; then $f'(x) = c_1 \exp(-2\mu x) \implies f(x) = \tilde{c}_1 \exp(-2\mu x) + \tilde{c}_2$,
 $f(a) = 0, f(b) = 1$.

Proposition 3.3.6

For Brownian motion with drift $\mu \neq 0$, for $a \leq x \leq b$:

$$P_x\{Z_{T_{a,b}} = b\} = \frac{\exp(-2\mu x) - \exp(-2\mu a)}{\exp(-2\mu b) - \exp(-2\mu a)}$$

Corollary 3.3.7

Let us have a Brownian motion with drift: $Z_t = B_t + \mu t$, $\mu < 0$.

Set $M = \sup_{t \in \mathbb{R}_+} Z_t$. Then for $m \geq 0$, $\mathbb{P}\{M \geq m\} = e^{2\mu m}$.

(And therefore $\mathbb{P}\{M = \infty\} = 0$, and $\mathbb{E}(M) = \int_0^\infty \mathbb{P}\{M \geq m\} dm = -\frac{1}{2\mu}$ (which is (> 0))).

Proof. Set $\tau_a = \inf\{t \geq 0 : Z_t = a\}$. By the previous proposition, for $-a < 0 < b$:

$$\mathbb{P}_0\{\tau_b < \tau_{-a}\} = \frac{\exp(-2\mu b) - \exp(2\mu a)}{\exp(-2\mu b) - \exp(2\mu a)}$$

Suppose that (**) $\mathbb{P}_0\{\lim_{a \rightarrow \infty} \tau_{-a} = \infty\} = 1$. Let $a \uparrow \infty$ in the equation above, then:

$$\mathbb{P}\{M \geq b\} = \mathbb{P}_0\{\tau_b < \infty\} = \frac{1 - 0}{\exp(-2\mu b) - 0} = e^{2\mu b}$$

Then all that remains to prove is (**). The map $a \rightarrow \tau_{-a}$ is nondecreasing (if $a_1 < a_2$, then (Z_t) must hit $-a_1$ before $-a_2$, so $\tau_{-a_1} < \tau_{-a_2}$. In particular, $\lim_{a \rightarrow \infty} \tau_{-a} = \sup_{a \in \mathbb{R}_+} \tau_{-a}$.

For $t > 0$, $\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t\} = \bigcap_{a \in \mathbb{N}} \{\tau_{-a} \leq t\} = \{\inf_{s \in [0, t]} Z_s = -\infty\}$.

(To get down to every level, you have to go down forever...)

Since $s \rightarrow Z_s(a)$ is continuous, then $\mathbb{P}_0\{\inf_{s \in [0, t]} Z_s = -\infty\} = 0 = P\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t\}$.

As t increases, $\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t\}$ increases, and so if $t_1 \leq t_2$, then $\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t_1\} \subset \{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t_2\}$ and so:

$$\mathbb{P}_0\left(\bigcup_{t \geq 0, t \in \mathbb{N}} \left\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t\right\}\right) = \lim_{t \rightarrow \infty, t \in \mathbb{N}} \mathbb{P}\left\{\sup_{a \in \mathbb{R}_+} \tau_{-a} \leq t\right\} = 0 \quad \text{so} \quad \mathbb{P}_0\left\{\sup_{a \in \mathbb{R}_+} \tau_{-a} = \infty\right\} = 1$$

□

Proposition 3.3.8

Asymptotic behaviour of $t \rightarrow Z_t$ (for $\mu < 0$).

We have $Z_t = B_t + \mu t$, $\mathbb{E}(B_t^2) = t$, so $B_t \sim \sqrt{t}$.

For t small ($t \approx 0$), $\sqrt{t} \gg t$, so Z_t behaves like B_t , but once t is large ($t \rightarrow \infty$), $\sqrt{t} \ll t$, then Z_t behaves like μt (drifts down to $-\infty$ at speed μ).

3.4 Zero sets

Definition 3.4.1: Zero set of a Brownian motion

Let $Z(\omega)$ denote the zero set; we write $Z(\omega) = \{t \in \mathbb{R}_+ : B_t(\omega) = 0\}$, for (B_t) a standard Brownian motion.

Recall that $P_0\{\exists t \in (t_0, t_1) : B_t = 0\} = \frac{2}{\pi} \arccos(\sqrt{\frac{t_0}{t_1}})$.

Proposition 3.4.2

The following is true:

$$\mathbb{P}_0\left(\bigcap_{n \in \mathbb{N}^*} \left\{ \exists t \in \left(0, \frac{1}{n}\right) : B_t = 0 \right\}\right) = 1.$$

(Therefore, there exists (t_n) such that $t_n > t_{n+1} > 0$, $\lim_{n \rightarrow \infty} t_n = 0$, and $B_{t_n} = 0$.)

Proof. Let $n \in \mathbb{N}^*$. Then:

$$\mathbb{P}_0\left\{ \exists t \in \left(0, \frac{1}{n}\right) : B_t = 0 \right\} \geq \mathbb{P}_0\left\{ \exists t \in \left(\frac{1}{m}, \frac{1}{n}\right) : B_t = 0 \right\}$$

(for all $m > n$)

$$= \frac{2}{\pi} \arccos\left(\sqrt{\frac{n}{m}}\right) \xrightarrow{m \rightarrow \infty} \frac{2}{\pi} \arccos(0) = 1.$$

Set $F_n = \left\{ \exists t \in \left(0, \frac{1}{n}\right) : B_t = 0 \right\}$. Then $F_n \supset F_{n+1}$ and therefore

$$\mathbb{P}_0\left(\bigcap_{n \in \mathbb{N}^*} \left\{ \exists t \in \left(0, \frac{1}{n}\right) : B_t = 0 \right\}\right) = \mathbb{P}_0\left(\bigcap_{n \in \mathbb{N}^*} F_n\right) = \lim_{n \rightarrow \infty} P_0(F_n) = 1.$$

□

Remark.

When the Brownian motion starts at $x_0 \neq 0$, set $\tau_0 = \inf\{t \geq 0 : B_t = 0\}$; then in $[t_0, t_0 + \varepsilon]$ there will be an infinite number of visits to 0.

Proposition 3.4.3

The following is true:

$$\mathbb{P}\left\{ \exists 0 \leq t_0 < t_1 : B_t = 0, \forall t \in [t_0, t_1] \right\} = 0$$

(Where the event above is that the Brownian motion stops at 0).

Proof. For $N \in \mathbb{N}$, define $F_N = \left\{ \exists 0 \leq t_0 < t_1 \leq N : B_t = 0, \forall t \in [t_0, t_1] \right\}$, then

Since $\mathbb{P}_0\left(\bigcup_{N \in \mathbb{N}} F_N\right) = \lim_{N \rightarrow \infty} P_0(F_N)$, it suffices to show that $\mathbb{P}_0(F_N) = 0, \forall N$. If F_N occurs, then:

$$\int_0^N \mathbf{1}_{\{B_t=0\}} dt \geq \int_{t_0}^{t_1} 1 dt = t_1 - t_0 > 0.$$

Set $Y = \int_0^N \mathbf{1}_{\{B_t=0\}} dt$, so $Y \geq 0$. Then:

$$E_0(Y) = E_0\left(\int_0^N \mathbf{1}_{\{B_t=0\}} dt\right) = \int_0^N E_0(\mathbf{1}_{\{B_t=0\}}) dt = \int_0^N P_0\{B_t = 0\} dt = 0$$

So $E_0(Y) = 0, \mathbb{P}\{Y \geq 0\} = 1 \implies \mathbb{P}\{Y > 0\} = 0$. Since $\{Y > 0\} \supset F_N, \mathbb{P}_0(F_N) = 0$, for all $N \in \mathbb{N}$.

Remark.

Fix $N \in \mathbb{N}$. Define $Z(\omega) = \{t \in (0, N) : B_t(\omega) = 0\}$.

Then $Z(\omega)$ is a closed set (in $(0, N)$) and $Z^c(\omega)$ is an open set and so $Z^c(\omega) = \bigcup_{i=1}^{\infty} \sigma_i$ (disjoint open intervals). Then $Z(\omega) \cup Z^c(\omega) = (0, N)$, and importantly, $\mathbf{1}_{Z(\omega)} =$

and so

$$\int_0^N \mathbf{1}_{Z(\omega)}(t) dt + \int_0^N \mathbf{1}_{Z^c(\omega)}(t) dt = 0 + N = N.$$

(Since the length of $Z(\omega) = 0$).

□

3.5 Hausdorff dimension

We move on to the Hausdorff dimension of $Z(\omega)$.

Definition 3.5.1: Cover

Fix $E \subset \mathbb{R}$. A family $(U_i, i \in \mathbb{N})$ of intervals is a cover of E if $E \subset \bigcup_{i \in \mathbb{N}} U_i$.

Definition 3.5.2: δ -cover

Let $L(U_i)$ be the length of an interval U_i . Fix $\delta > 0$. (U_i) is a δ -cover of E if it is a cover and $L(U_i) \leq \delta$.

Fix $\alpha > 0$. Define $\mu_\alpha(E) = \lim_{\delta \downarrow 0} \inf_{U_i \delta\text{-cov. of } E} \sum_{i=1}^{\infty} (L(U_i))^\alpha$, and so $0 \leq \mu_\alpha \leq \infty$.

Lemma 3.5.3: Hausdorff construction process lemma

- a) Fix $\alpha > 0$. Suppose that $\mu_\alpha(E) < \infty$. Then for $\beta > \alpha$, $\mu_\beta(E) = 0$.
- b) Fix $\alpha > 0$. Suppose that $\mu_\alpha(E) > 0$. Then for $\beta < \alpha$, $\mu_\beta(E) = \infty$.

Corollary 3.5.4

As a consequence, $\exists \alpha_0$ such that $\beta > \alpha_0 \implies \mu_\beta(E) = 0$, and $\beta < \alpha_0 \implies \mu_\beta(E) = \infty$. Then $\mu_{\alpha_0}(E)$ can be 0, ∞ , or in $(0, \infty)$.

α_0 is called the Hausdorff dimension of E .

Definition 3.5.5: Hausdorff dimension in $\mathbb{R}^d$

Let $E \subset \mathbb{R}^d$. (U_i) is a δ -cover of E if $E \subset \bigcup_i U_i$, and each U_i is a ball with radius $\varepsilon_i \geq 0$, and $\text{diam} U_i = 2\varepsilon_i \leq \delta$.

($U_i = \emptyset$ is allowed, Euclidean norm or any other norm, balls can be open or closed.)

For $\alpha > 0$, $\mu_\alpha(E) = \lim_{\delta \downarrow 0} \inf_{U_i \delta\text{-cov. of } E} \sum_{i=1}^{\infty} (L(U_i))^\alpha$. Then

$$\dim_H(E) = \inf\{\alpha > 0 : \mu_\alpha(E) = 0\} = \sup\{\alpha > 0 : \mu_\alpha(E) = \infty\}$$

Theorem 3.5.6

Let (B_t) be a standard Brownian motion. Then:

$$\mathbb{P}_0 \left\{ w : \dim_H \{t \in \mathbb{R}_+ : B_t(w) = 0\} = \frac{1}{2} \right\} = 1.$$

The proof requires some results.

Proposition 3.5.7

$$\mathbb{P}_0 \left\{ w : \dim_H \{t \in \mathbb{R}_+ : B_t(w) = 0\} \leq \frac{1}{2} \right\} = 1.$$

Lemma 3.5.8

$\exists 0 < c_0 < c_1 < \infty$ and $s_0 > 0$ such that for all $t \in [1, 2]$ and $0 \leq s \leq s_0$:

$$c_0 \sqrt{s} \leq \arccos \sqrt{\frac{t}{t+s}} \leq c_1 \sqrt{s}$$

(Proof of the lemma is an exercise in analysis; write $y = \frac{t+s}{t}$, do some rearranging, show that $\lim_{y \downarrow 1} \frac{\arccos(\frac{1}{\sqrt{y}})}{\sqrt{y-1}} = 1$ by L'Hopital's.)

Lemma 3.5.9

For $I = [t_0, t_1] \subset [1, 2]$, $t_0 < t_1$, $t_1 - t_0$ small enough:

$$\mathbb{P}_0 \{ \exists t \in I : B_t = 0 \} \leq \frac{2}{\pi} c_1 \sqrt{L(I)}$$

Proof. (Of the theorem.) Let $Z(w) = \{t \in [1, 2] : B_t(w) = 0\}$. Define a cover:

$$U_i^n(w) = \begin{cases} [1 + \frac{i}{n}, 1 + \frac{i+1}{n}] & \text{if } Z(w) \cap [1 + \frac{i}{n}, 1 + \frac{i+1}{n}] \neq \emptyset \\ \emptyset & \text{otherwise} \end{cases}$$

Suppose (as an example) the Brownian motion crosses 0 at around (at least) three certain points [he drew a picture]. Then the corresponding intervals would be $U_1^n = [1 + \frac{1}{n}, 1 + \frac{2}{n}]$, $U_4^n = [1 + \frac{4}{n}, 1 + \frac{5}{n}]$, $U_7^n = [1 + \frac{7}{n}, 1 + \frac{8}{n}]$, $U_j^n = \emptyset, j \notin \{1, 4, 7, \dots\}$.

Clearly (vacating the example), $Z(w) \subset \bigcup_{i=0}^{n-1} U_i^n$, so $(U_i^n, i = 0, \dots, n-1)$ is a $\frac{1}{n}$ -cover of Z . Fix $\alpha > \frac{1}{2}$. We will try to show that $\mathbb{P}\{\mu_\alpha(Z) = 0\} = 1$.

Now $\mu_\alpha \geq 0$:

$$0 \leq E_0(\mu_\alpha(Z)) \leq E_0 \left(\liminf_{n \rightarrow \infty} \sum_{i=0}^{n-1} (L(U_i^n))^\alpha \right)$$

Which by Fatou's lemma:

$$\begin{aligned} &\leq \liminf_{n \rightarrow \infty} E_0 \left(\sum_{i=0}^{n-1} (L(U_i^n))^\alpha \right) = \liminf_{n \rightarrow \infty} \sum_{i=0}^{n-1} E_0((L(U_i^n))^\alpha) \\ &= \liminf_{n \rightarrow \infty} \sum_{i=0}^{n-1} E_0 \left(\left(\frac{1}{n} \right)^\alpha \mathbf{1}_{\{U_i^n \neq \emptyset\}} + 0^\alpha \mathbf{1}_{\{U_i^n = \emptyset\}} \right) \\ &= \liminf_{n \rightarrow \infty} \sum_{i=0}^{n-1} \left(\frac{1}{n} \right)^\alpha \mathbb{P}_0 \{ \exists t \in [1 + \frac{i}{n}, 1 + \frac{i+1}{n}] . B_t = 0 \} \end{aligned}$$

By Lemma 2 (above):

$$\leq \liminf_{n \rightarrow \infty} \sum_{i=0}^{n-1} \left(\frac{1}{n} \right)^\alpha c \sqrt{\frac{1}{n}} = \liminf_{n \rightarrow \infty} n^{1-\alpha-\frac{1}{2}} = \liminf_{n \rightarrow \infty} n^{\frac{1}{2}-\alpha} = 0$$

Therefore, $E_0(\mu_\alpha(Z)) = 0 \implies \mathbb{P}_0\{\mu_\alpha(Z) = 0\} = 1 \implies \mathbb{P}_0\{\dim_H(Z) \leq \frac{1}{2}\} = 1$. $\square$

Proof. Extension to $\mathbb{R}_+$:

We have proven it for $Z(w)$ restricted to $[1, 2]$. Now let $Z(w) = \{t \in \mathbb{R}_+ : B_t(w) = 0\}$. We have just seen that for $\alpha > \frac{1}{2}$, $P_0\{\mu_\alpha(Z \cap [1, 2]) = 0\} = 1$.

Similarly (for $\alpha > \frac{1}{2}$) $\mathbb{P}_0\{\mu_\alpha(Z \cap [n, n+1]) = 0\} = 1, \forall n \geq 1$, and $P_0\{\mu_\alpha(Z \cap (\frac{1}{n+1}, \frac{1}{n}]) = 0\} = 1, \forall n \geq 1$. Therefore:

$$\mu_\alpha(Z) = \mu_\alpha \left(\{0\} \cup \bigcup_{n=1}^{\infty} \left(Z \cap \left(\frac{1}{n+1}, \frac{1}{n} \right] \right) \cup \left(\bigcup_{n=1}^{\infty} (Z \cap [n, n+1]) \right) \right)$$

(disjoint union, μ_α is a measure)

$$= \mu_\alpha(\{0\}) + \sum_{n=1}^{\infty} \mu_\alpha(Z \cap (\frac{1}{n+1}, \frac{1}{n}]) + \sum_{n=1}^{\infty} \mu_\alpha(Z \cap [n, n+1)) = 0 \text{ with prob. } 1.$$

Therefore $\mathbb{P}_0\{\mu_\alpha(Z) = 0\} = 1 \forall \alpha > \frac{1}{2}$ and so $P_0\{\dim_H(Z) \leq \frac{1}{2}\} = 1$. $\square$

3.6 Continuity of sample paths

Objective: We want to construct a random continuous function $t \rightarrow B_t(w)$ which satisfies properties (a) and (b) of the definition of a BM.

Paul Lévy constructed an idea: Let (X_n) be a sequence of i.i.d $N(0, 1)$ r.v.s. Define $S_0 = 0$, and for $n \geq 1$, $S_n = X_1 + \dots + X_n$. Then (S_n) is a Gaussian random walk on $\mathbb{R}$.

We view (S_n) as the restriction to $\mathbb{N}$ of a Brownian motion on $\mathbb{R}_+$. Then we have the Brownian motion at integer times. We must define B_t for $t \in \mathbb{R}_+ \setminus \mathbb{N}$.

Step 1. Linear interpolation:

Define:

$$B_t^{(1)} = \begin{cases} B_t & \text{if } t \in \mathbb{N} \\ B_n + (t - n)(B_{n+1} - B_n) & \text{if } t \in [n, n+1], n \in \mathbb{N} \end{cases}$$

Then, by exercise 3 on problem set 8:

$$B_{n+\frac{1}{2}}^{(1)} = \frac{B_n^{(1)} + B_{n+1}^{(1)}}{2} = \mathbb{E}(B_{n+\frac{1}{2}} \mid B_n, B_{n+1})$$

Recall that given B_n, B_{n+1} , the law of $B_{n+\frac{1}{2}}$ is $N(\frac{B_n+B_{n+1}}{2}, \frac{1}{4})$

Step 2. Refine the time step:

Define $B_n^{(2)} = B_n^{(1)} = S_n, n \in \mathbb{N}$. Then $B_{n+\frac{1}{2}}^{(2)} = \frac{B_n^{(1)} + B_{n+1}^{(1)}}{2} + \frac{1}{2}Z_n^{(1)}, n \in \mathbb{N}$, where $(Z_n^{(1)}, n \in \mathbb{N})$ are i.i.d $N(0, 1)$, and independent of (X_n) .

Define $B_{\frac{m}{2}}^{(2)} = B_{\frac{m}{2}}^{(1)}, m \in \mathbb{N}$, so $B_{\frac{2m}{2}}^{(2)} = B_m^{(2)} = B_m^{(1)}, m \in \mathbb{N}$.

(So now we've gone to all the half-integers and added a (by random variable) point there, and interpolated in between, and then we'll go in between all the half-integers and the integers...)

Construction on $[0, 1]$:

For $n \in \mathbb{N}$, set $D_n = \{\frac{k}{2^n}, k = 0, 1, \dots, 2^n\}$ (dyadics of order n in $[0, 1]$), so $D_0 = \{0, 1\}$, $D_1 = \{0, \frac{1}{2}, 1\}$, $D_2 = \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}, \dots$

So $D_n \subset D_{n+1}$. Let $D = \bigcup_{n \in \mathbb{N}} D_n$: the dyadics in $[0, 1]$. Then D is countable and dense in $[0, 1]$.

Fix $(Z_d, d \in D)$ as i.i.d $N(0, 1)$ r.v.s. We begin by defining B_t for $t \in D$.

For $n = 0$: set $B_0 = 0, B_1 = Z_1$. This defines B_t for $t \in D_0$.

For $n \geq 1$, we will define inductively B_t for $t \in D_n$, such that:

- i) for $r < s < t$ in D_n , $B_t - B_s$ in $N(0, t - s)$ and independent of $B_s - B_r$.
- ii) for $t \in D_n$, B_t is determined by the $Z_e, e \in D_n$ (and therefore is independent of the $Z_e, e \in D \setminus D_n$),

Clearly for $n = 0$, the two properties hold. For $n \geq 1$, suppose, by induction, that $(B_t, t \in D_{n-1})$ has been defined in such a way that i) and ii) hold for $n - 1$.

For $d \in D_n \setminus D_{n-1}$, define $d_n^- = d - 2^{-n}$, $d_n^+ = d + 2^{-n}$. Then $d^-, d^+ \in D_{n-1}$. Define:

$$B_d = \frac{1}{2}(B_{d_n^-} + B_{d_n^+}) + \frac{1}{2^{\frac{n+1}{2}}}Z_d \quad (**)$$

So then $(B_t, t \in D_n)$ is defined. We need to check that i) and ii) are satisfied. Observe that $X_1 = \frac{1}{2}(B_{d_n^-} - B_{d_n^+})$, by induction, is $N(0, \frac{1}{2^2} \frac{2}{2^n}) = N(0, \frac{1}{2^{n+1}})$. Then $X_2 = \frac{1}{2^{\frac{n+1}{2}}}Z_d$ is $N(0, \frac{1}{2^{n+1}})$ as well.

By ii) for $n - 1$, X_1 and X_2 are independent, therefore their sum and difference are independent and $N(0, \frac{1}{2^{n+1}} + \frac{1}{2^{n+1}}) = N(0, \frac{1}{2^n})$.

(True by computing covariance: $\mathbb{E}((X_1 + X_2)(X_1 - X_2)) = \mathbb{E}(X_1^2) - \mathbb{E}(X_2^2) = 0$.)

Moreover, $X_1 + X_2 = B_d - B_{d_n^-}$, and $X_1 - X_2 = \frac{1}{2}(B_{d_n^+} - B_{d_n^-}) - \frac{1}{2^{\frac{n+1}{2}}}Z_d = B_{d_n^+} - B_d$.

So as a consequence, for $d < f$ where $d, f \in D_n \setminus D_{n-1}$, $B_d - B_{d_n^-}, B_{d_n^+} - B_{d_n^-}, B_f - B_{f_n^-}, B_{f_n^+} - B_f$ are independent pairs, because we have just seen that the first two are independent. The same holds (for the second pair), and for example, if we want, $B_{d_n^+} - B_d$ and $B_f - B_{f_n^-}$ are also independent because they are determined by r.v.s that are independent of each other.

Conclusion.

$(B_d, d \in D)$ satisfies property i) because an increment over a long interval is the sum of increments over intervals of length 2^{-n} . It also satisfies ii) by construction (especially by equality (**)).

So $(B_d, d \in D)$ has the properties of Brownian motion.

Extension to $[0, 1] \setminus D$:

Define $F_0(t) = tZ_1, t \in [0, 1]$, and define, for $n \geq 1$:

$$F_n(t) = \begin{cases} 2^{-\frac{n+1}{2}}Z_t & \text{if } t \in D_n \setminus D_{n-1} \\ 0 & \text{if } t \in D_{n-1} \\ \text{linear (affine)} & \text{between consecutive elements of } D_n \end{cases}$$

Each F_n is a continuous function on $[0, 1]$. Now define $B_t^{(n)} = \sum_{i=0}^n F_i(t)$. This function is

continuous and affine between consecutive elements of D_n .

If $t = d \in D_{n-1}$, then $F_n(t) = 0$ and so $B_t^{(n)} = B_t^{(n-1)}$.

If $t \in D_n$ and $m > n$, then $B_t^{(m)} = B_t^{(n)}$ because for $n < i \leq m$, $F_i(t) = 0$ since $t \in D_n \subset D_{i-1}$.

Therefore, $B_t^{(n)} = \sum_{i=0}^n F_i(t) = \sum_{i=0}^{\infty} F_i(t)$, $t \in D_n$.

Definition 3.6.1

For $t \in D = \bigcup_{n=0}^{\infty} D_n$, let $\tilde{B}_t = \sum_{i=0}^{\infty} F_i(t)$ [so $\tilde{B}_t = B_t^{(n)}$ if $t \in D_n$.]

Claim

For $t = d \in D_n$, $\tilde{B}_t = B_d$ (from (**)).

Proof for Claim.

For $n = 0$: if $t = 0$, when $\tilde{B}_0 = 0 = B_0$. If $t = 1$, $\tilde{B}_1 = F_0(1) = Z_1 = B_1$.

Suppose the claim is true for $n - 1$. Then, by induction in the equality: for

$t \in D_n \cap D_{n-1}$, then $\tilde{B}_t = B_t$.

For $t = d \in D_n \setminus D_{n-1}$: for $i \in \{0, 1, \dots, n-1\}$, F_i is affine on $[d_n^-, d_n^+]$.

F_i must vanish at either $\frac{k}{2^i}$ or $\frac{k+1}{2^i}$.

Therefore, $\sum_{i=0}^{n-1} F_i(d) = \sum_{i=0}^{n-1} \frac{1}{2} (F_i(d_n^-) + F_i(d_n^+)) = \frac{1}{2} (\sum_{i=0}^{n-1} F_i(d_n^-) + \sum_{i=0}^{n-1} F_i(d_n^+))$ by induction $= \frac{1}{2} (B_{d_n^-} + B_{d_n^+})$.

Therefore $\tilde{B}_t = \sum_{i=0}^n F_i(d) = \frac{1}{2} (B_{d_n^-} + B_{d_n^+}) + F_n(d) = B_d$.

Claim

The series of functions $\sum_{n=0}^{\infty} F_n$ converges uniformly on $[0, 1]$ (with probability 1).

This also implies that $t \rightarrow \tilde{B}_t = \sum_{n=0}^{\infty} F_n(t)$ is a continuous function, w/ prob. 1.

Proof for Claim.

Notice that:

$$\sup_{t \in [0, 1]} |F_n(t)| \leq 2^{-\frac{n+1}{2}} \sup_{t \in D_n \setminus D_{n-1}} |Z_t|$$

$$\mathbb{P} \left\{ \sup_{t \in D_n \setminus D_{n-1}} |Z_t| \geq c\sqrt{n} \right\} \leq \mathbb{P} \left\{ \sup_{t \in D_n} |Z_t| \leq c\sqrt{n} \right\}$$

$$\leq \sum_{t \in D_n} \mathbb{P}\{|Z_t| \geq c\sqrt{n}\} = o$$

Now: Recall that if $Z \sim N(0, 1)$ and $x \geq 0$, we can bound: $\mathbb{P}\{|Z| \geq x\} \leq e^{-\frac{x^2}{2}}$. We use this:

$$o \leq (2^n + 1)e^{-\frac{c^2 n}{2}} \leq 2 \cdot 2^n e^{-\frac{c^2 n}{2}} = 2 \exp(n(\ln 2 - \frac{c^2}{2}))$$

If $\ln 2 - \frac{c^2}{2} < 0$ (i.e. $c > \sqrt{2 \ln 2}$, then $\sum_{n=0}^{\infty} \mathbb{P}\{\sup_{t \in D_n \setminus D_{n-1}} |Z_t| \geq c\sqrt{n}\} < \infty$.

Since we have a sum of finite probabilities, we can use the Borel-Cantelli lemma:

$\exists N(w)$ such that, for $n \geq N(w)$, $\sup_{t \in D_n \setminus D_{n-1}} |Z_t| < c\sqrt{n}$ and

$$\sum_{n=0}^{\infty} \sup_{t \in [0,1]} |F_n(t)| \leq c \sum_{n=0}^{\infty} \sqrt{n} 2^{-\frac{n+1}{2}} < \infty$$

Therefore $\sum_{n=0}^{\infty} F_n$ converges normally, and therefore uniformly on $[0, 1]$.

In conclusion, $t \rightarrow \tilde{B}_t = \sum_{i=0}^{\infty} F_i(t)$, $t \in [0, 1]$, defines a continuous function. It has properties a) and b) of Brownian motion, because $\tilde{B}_t = B_t$, $t \in D$, $(B_d, d \in D)$ have properties a) and b), and D is dense in $[0, 1]$ and $t \rightarrow \tilde{B}_t$ is continuous.